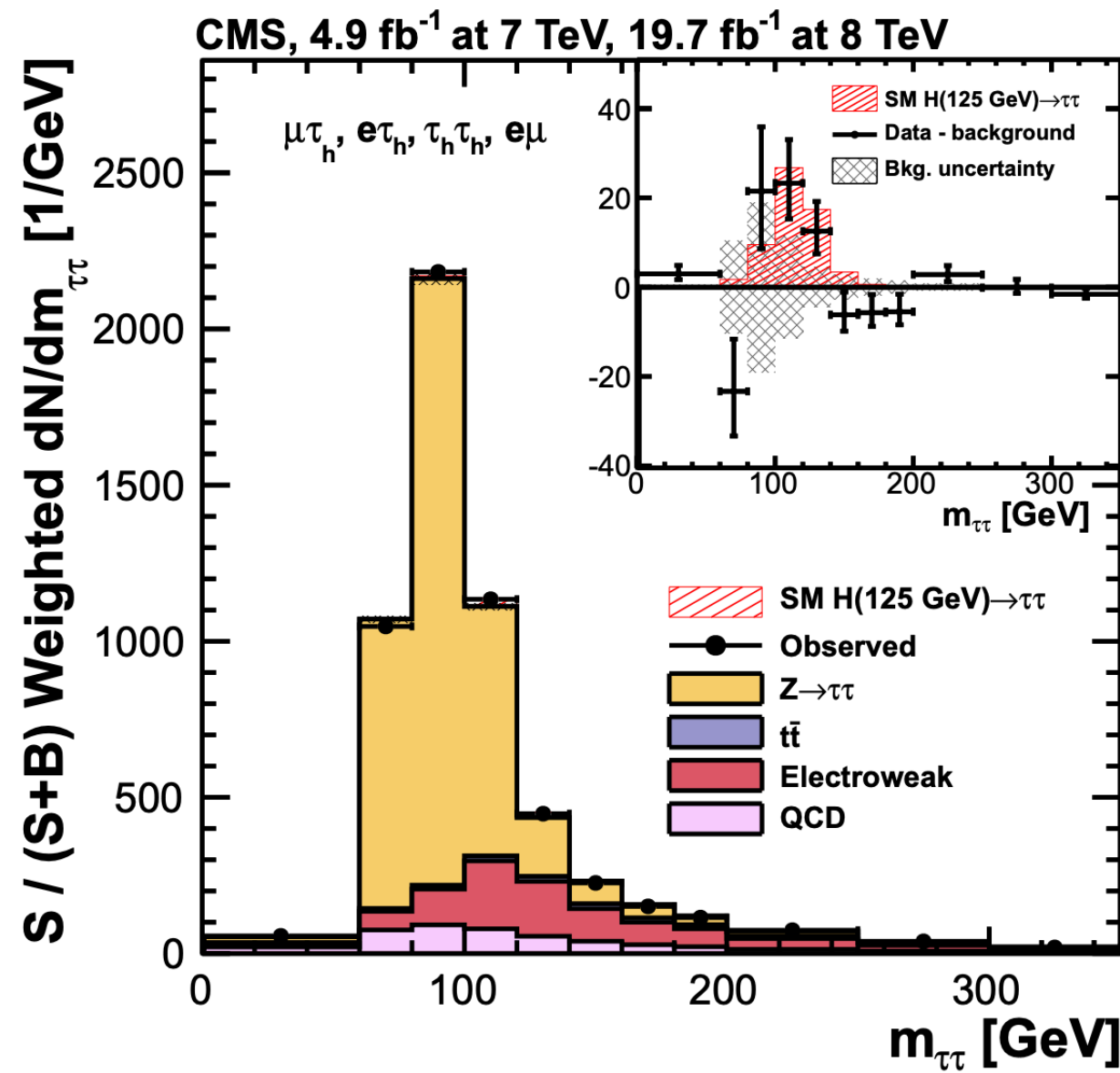




Lecture 14: Deep Learning Regressions

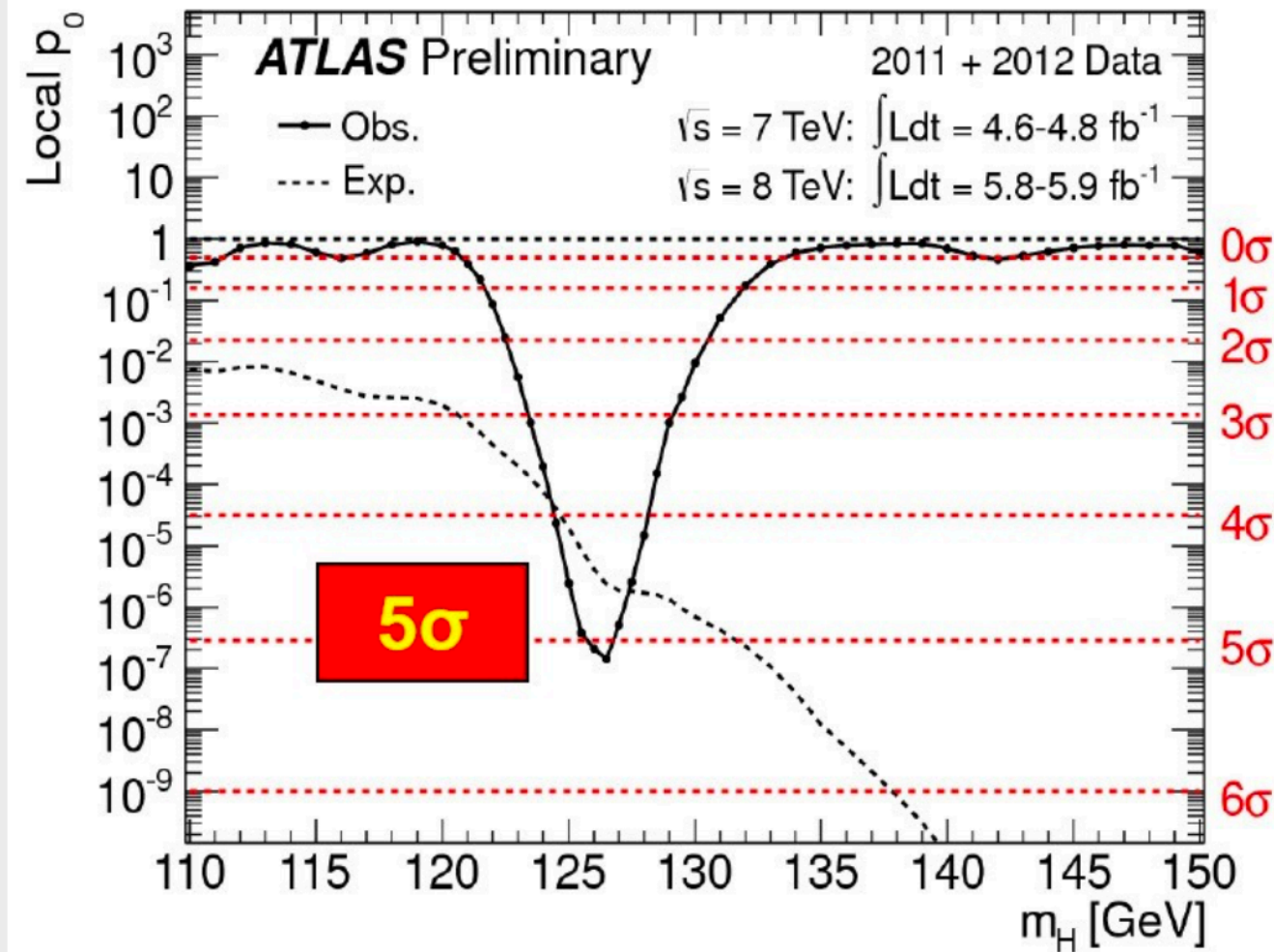
What you may not know?



At the Higgs discovery

ATLAS

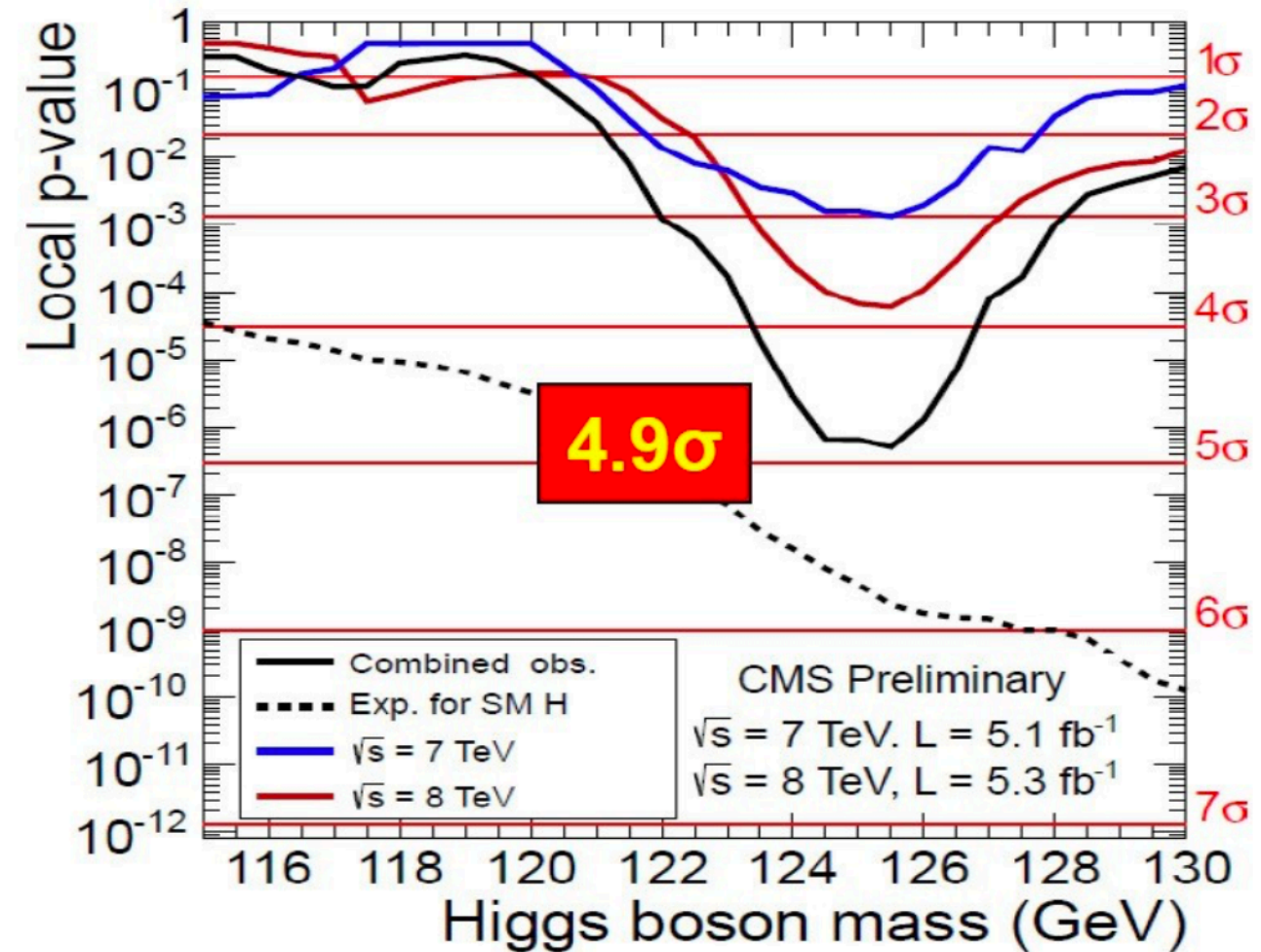
$\gamma\gamma, 4l$ updated with
 $\sim 6 \text{ fb}^{-1}$ of 8 TeV data



Largest local excess:
5 σ at $m_H = 126.5 \text{ GeV}$

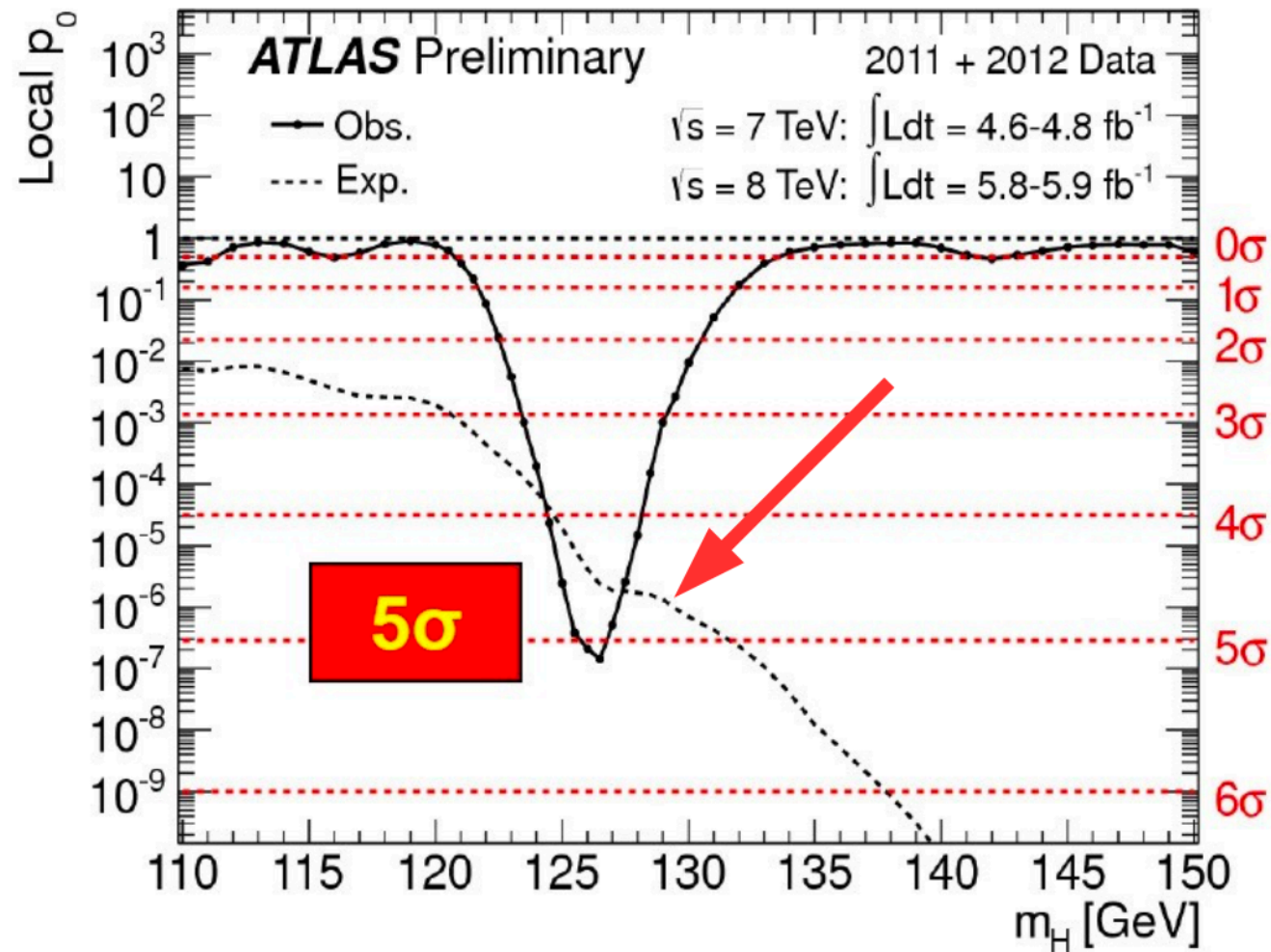
CMS

All channels updated with
 $\sim 5 \text{ fb}^{-1}$ of 8 TeV data

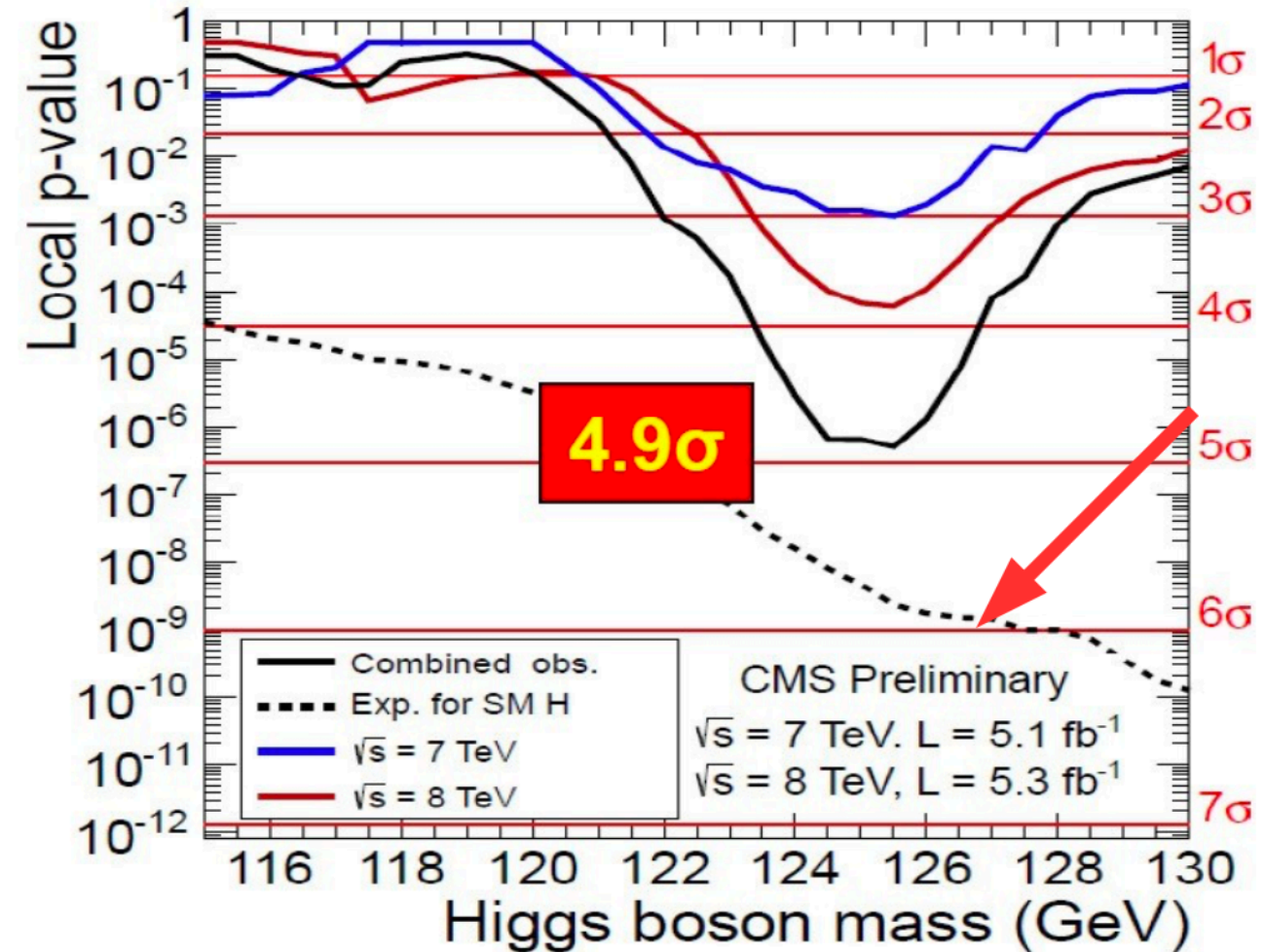


Largest local excess:
4.9 σ around $m_H = 125 \text{ GeV}$
(using $H \rightarrow \gamma\gamma$ and $H \rightarrow 4l$: 5.0 σ)

A big difference was present

ATLAS *$\gamma\gamma, 4l$ updated with
 $\sim 6 \text{ fb}^{-1}$ of 8 TeV data*

Largest local excess:
5 σ at $m_H = 126.5 \text{ GeV}$

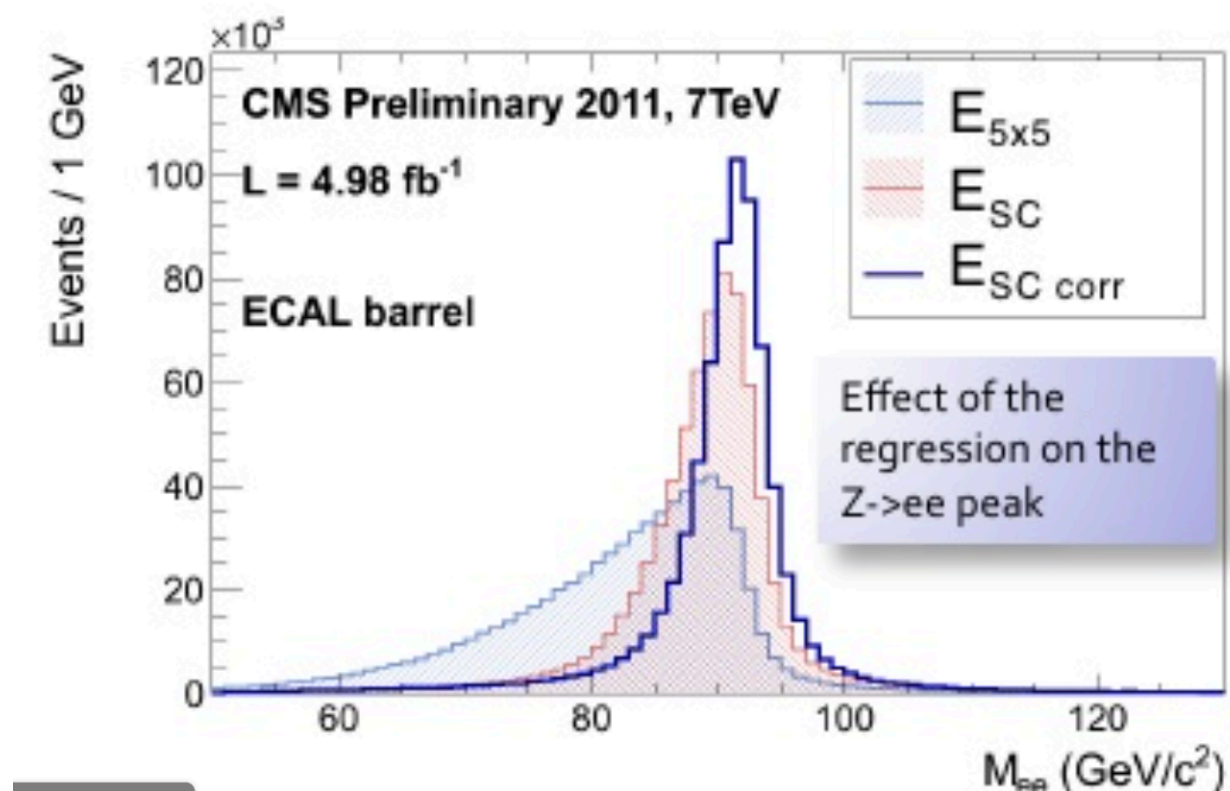
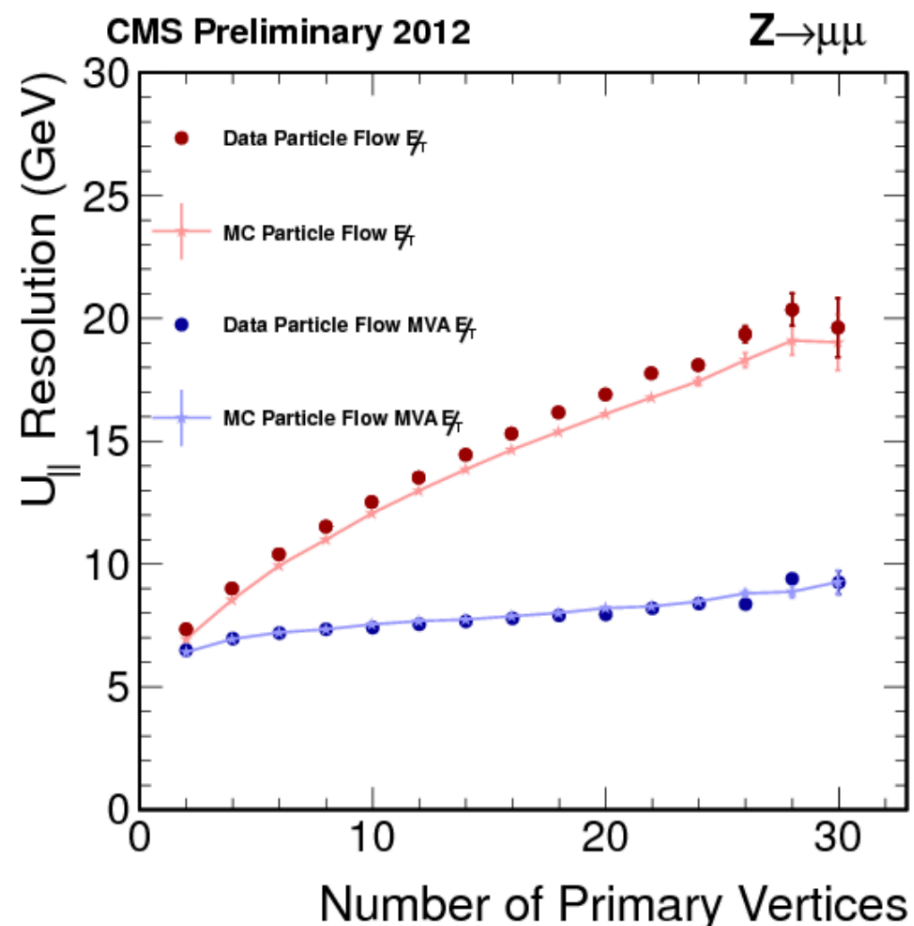
CMS*All channels updated with
 $\sim 5 \text{ fb}^{-1}$ of 8 TeV data*

Largest local excess:
4.9 σ around $m_H = 125 \text{ GeV}$
(using $H \rightarrow \gamma\gamma$ and $H \rightarrow 4l$: 5.0 σ)

**CMS was nearly 30% more sensitive
 Despite an excess of same size**

What caused the⁵ difference?

- A few things, but the big one was deep learning
- In particular, two novel deep learning approaches
 - These approaches involved deep learning regression



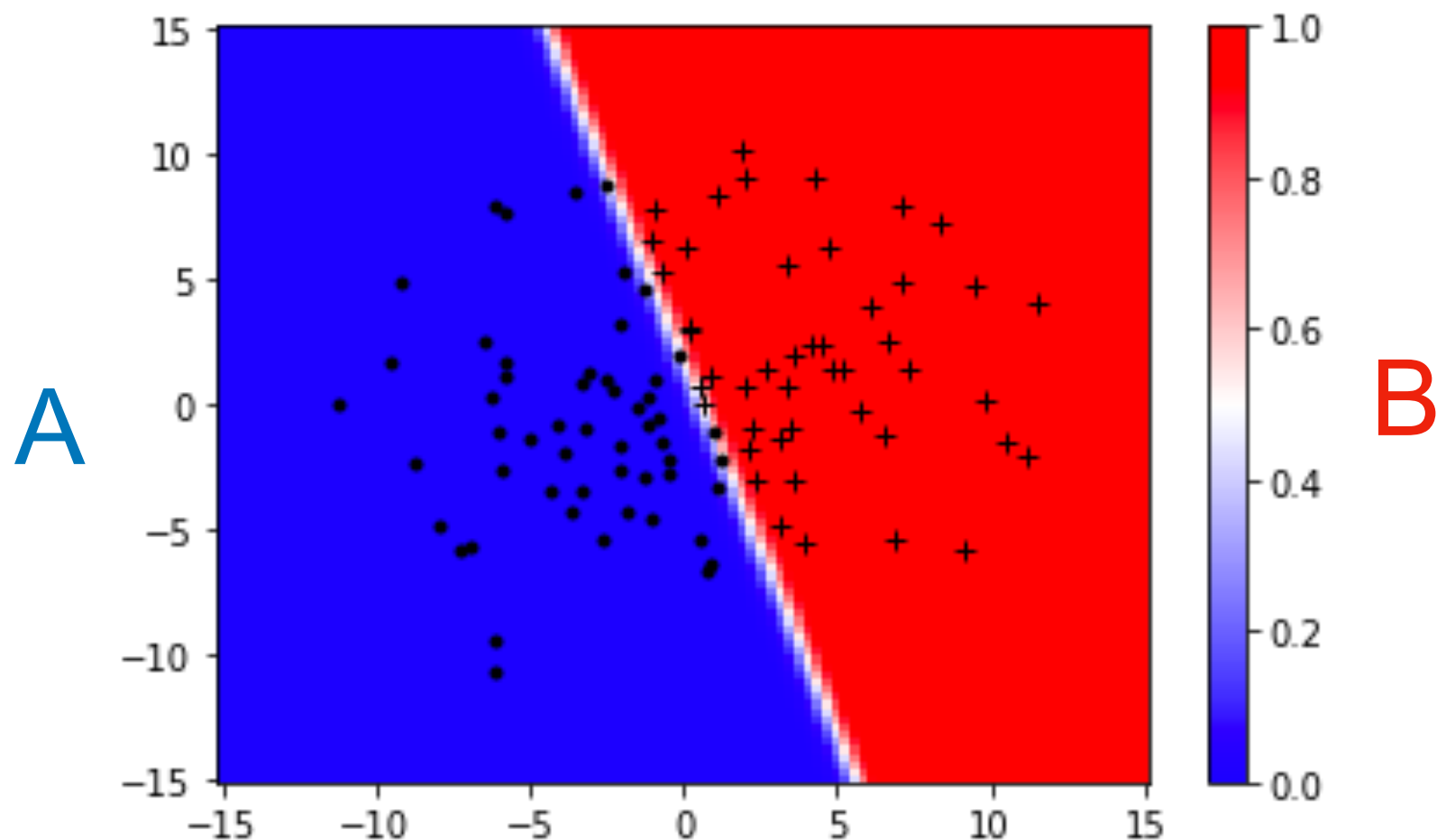
Overview

- In this lecture we are going to talk about
 - Deep Learning Regression
- Regression uses all the usual deep learning tools
 - Tries to solve a different problem than other DL lecture
 - Additionally it combines many of the concepts in fitting
- Lets review previous lectures to understand

Deep Learning

- In the past lectures we focused on :
 - Deep learning based classification

How do I separate two classes of points?



Deep Learning

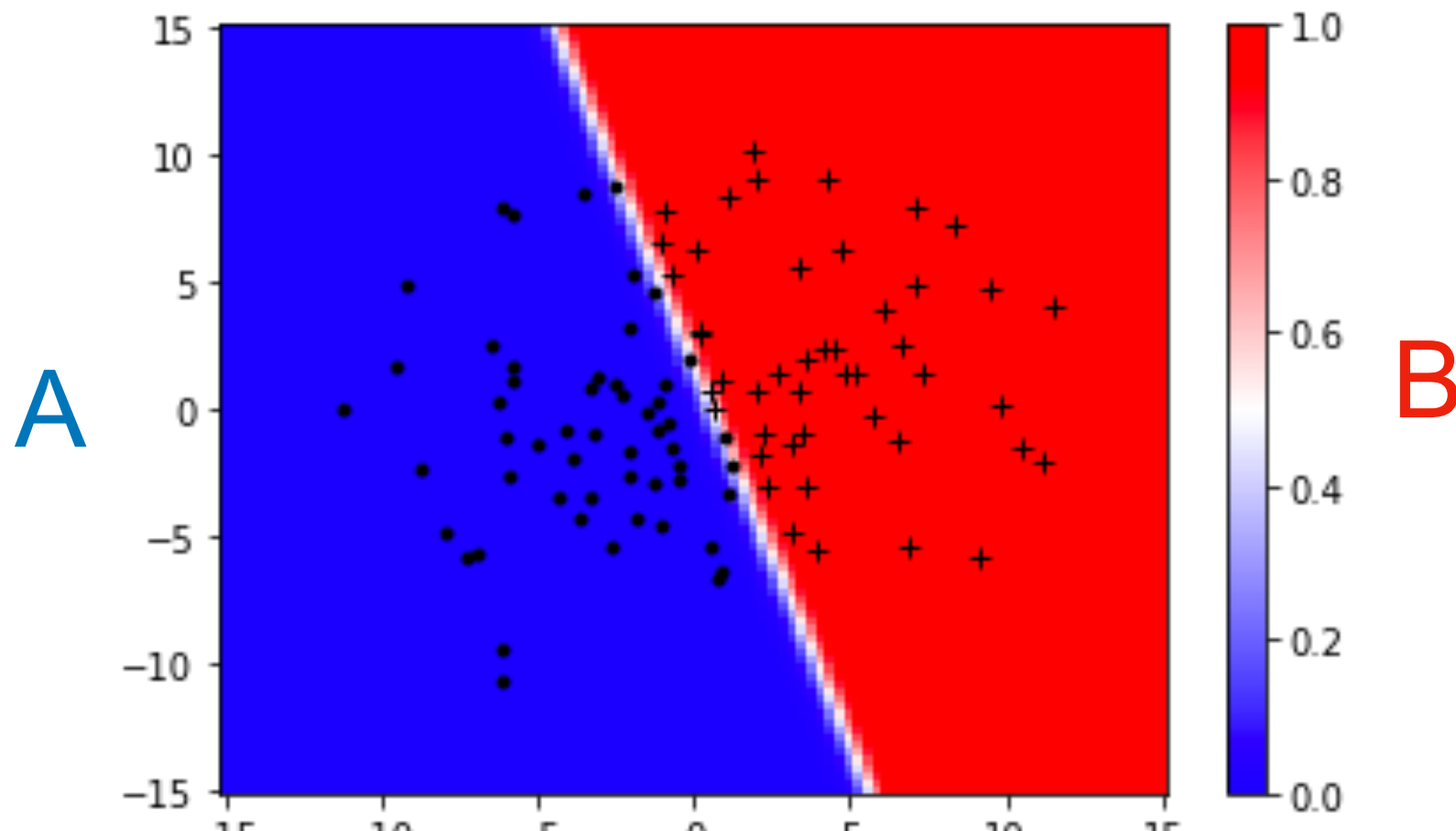
- In the past lectures we focused on :
- Deep learning based classification

How do I separate to classes of points?

Minimize Loss:

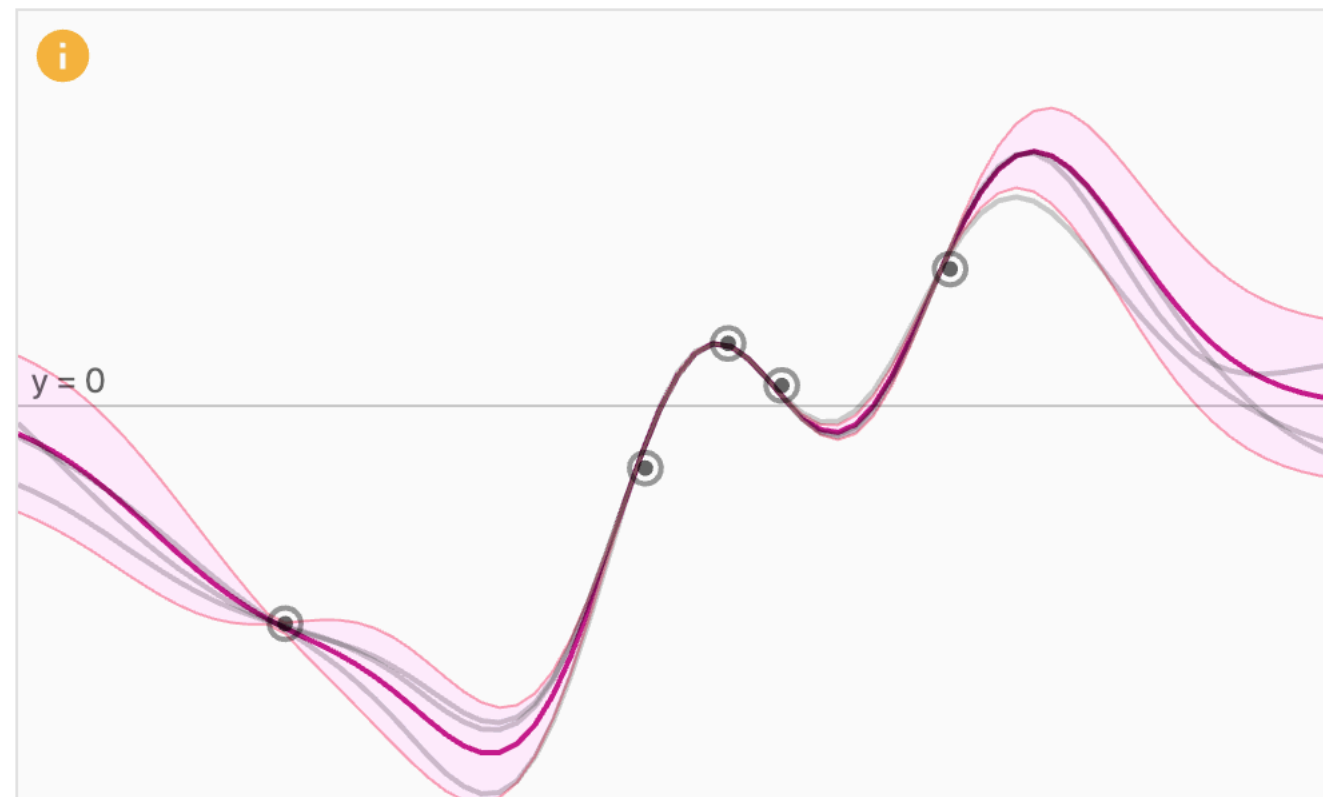
$$\mathcal{L} = B_{true} \log(p(B)) + A_{true} \log(p(A))$$

$$\mathcal{L} = (1 - A_{true}) \log(1 - p(A)) + A_{true} \log(p(A))$$



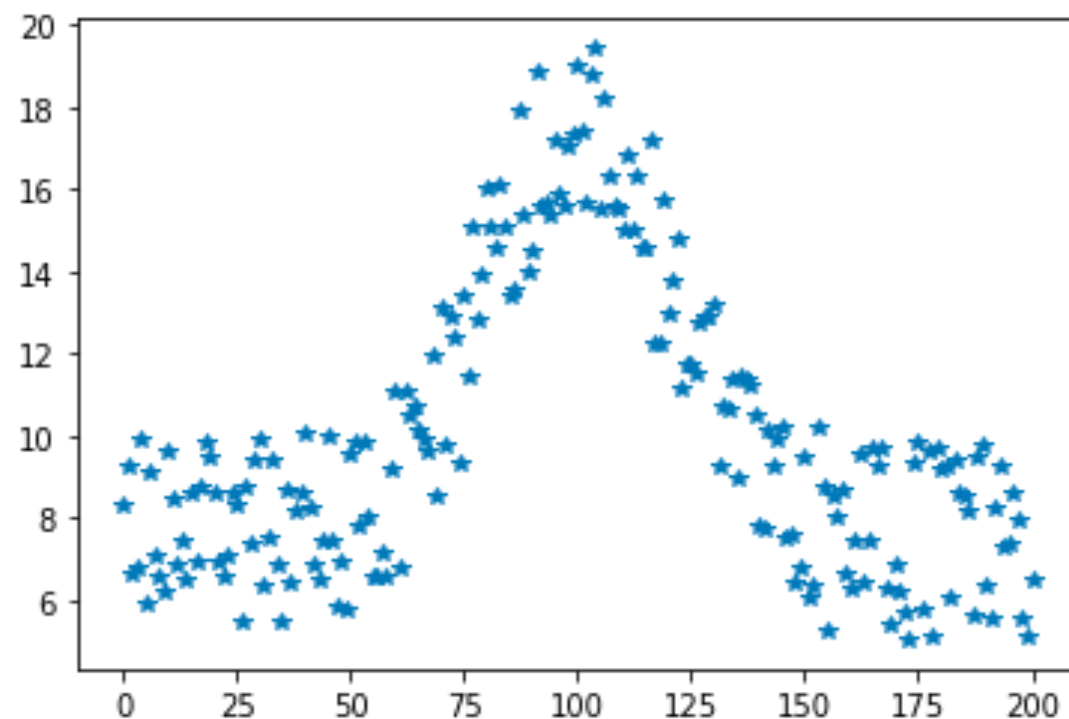
Interpolation

- How do I take a continuous set of points and connect them?
- We have considered two separate approaches
 - Fitting a range of polynomials
 - Spline Interpolation and Gaussian Processes



Fitting Any Distribution

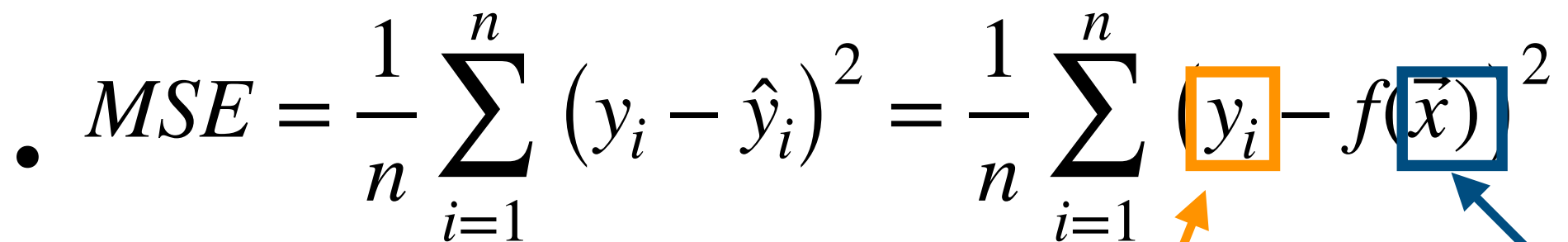
- Between minimizing the likelihood and statistics we know what to do to get a fit that describes the data well. With interpolation and gaussian processes, we can connect the dots. However there are limitations what if we want to do something more complicated!
- **Challenge:** Fit the points below without guessing a function.



To the notebook

How do we do w/NN?

- With an NN all we are doing is a minimizing a loss
 - This loss can be any loss in the end
 - Really **Whatever we want!**
- A common loss is so-called Mean Squared Error

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum_{i=1}^n (y_i - f(\vec{x}))^2$$


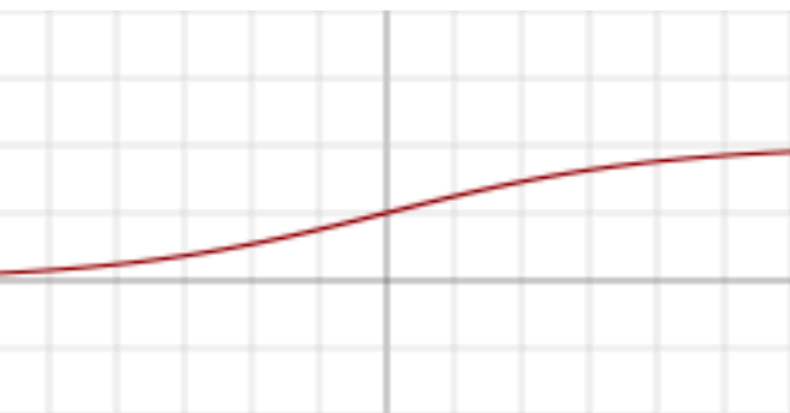
This is our input to our Neural Network it can be a vector of arbitrary size

This our target data in the training it can also be a vector of arbitrary size

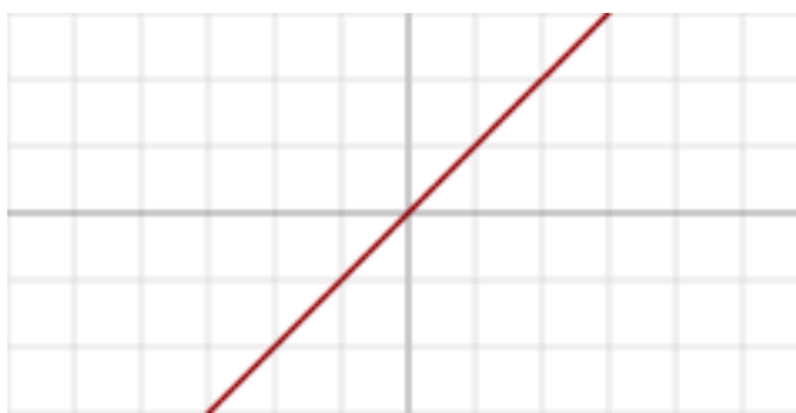
To the Notebook

Activation Functions

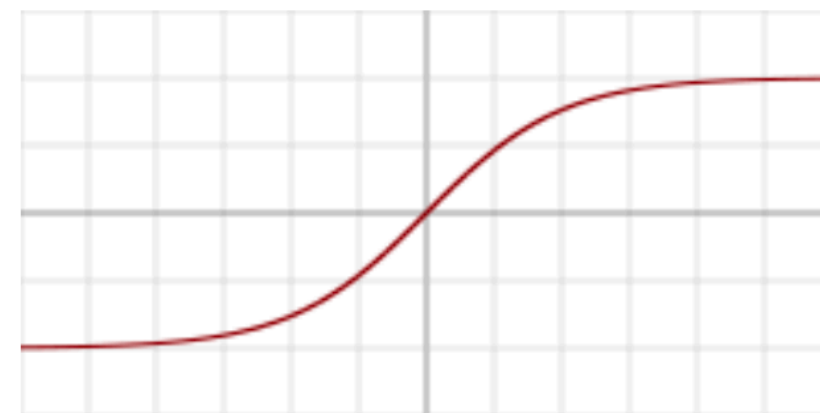
Sigmoid



Linear



Tanh



Softmax

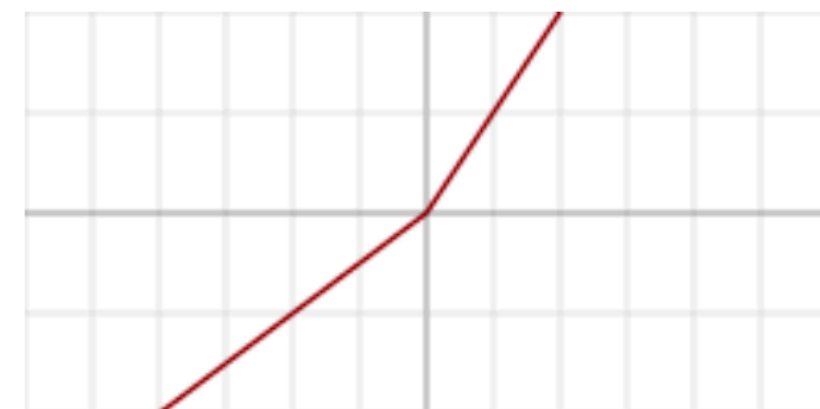
(multiclass)

$$\frac{e^{x_i}}{\sum_{j=1}^J e^{x_j}}$$

ReLU

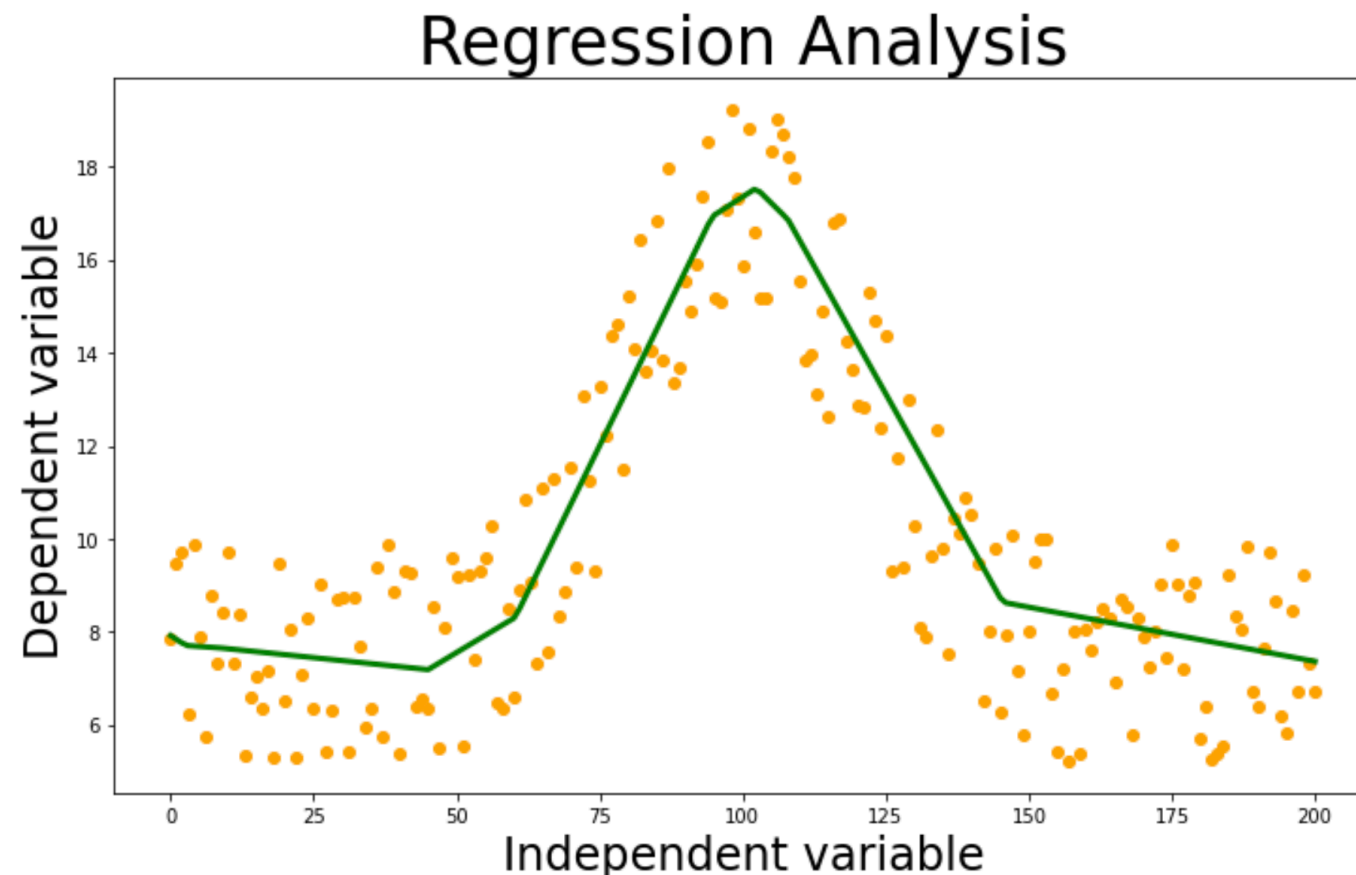


LeakyReLU



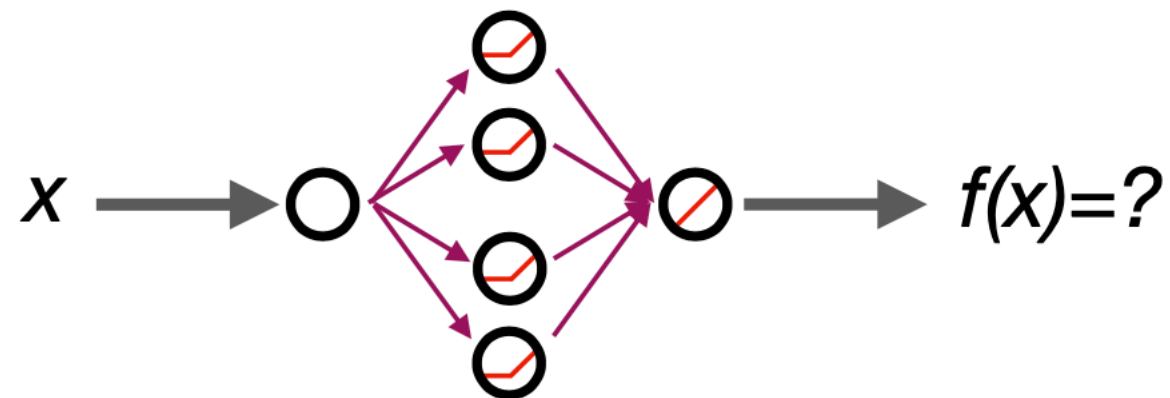
Parameter Extraction

- Despite being able to fit such a distribution
- There is a limit to how much we can do
- The functional form for this distribution is complicated
- To get a mean and a resolution, requires reverse engineering



Universal Functions

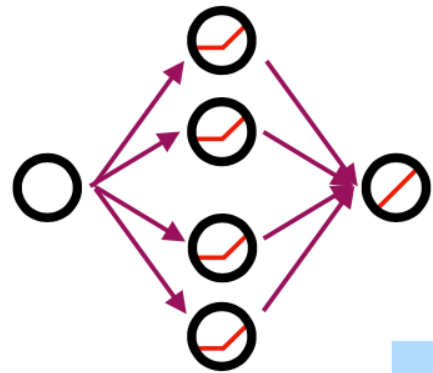
Approximating (Example)



What kind of function can this network architecture approximate?

Universal Functions

Approximating (Example)



$$= \tilde{\mathbf{W}}^T \mathbf{x} + \tilde{\mathbf{b}}, \quad \mathbf{W}_1^T \mathbf{x} + \mathbf{b}_1 > 0$$

$$\mathbf{b}_2, \quad \mathbf{W}_1^T \mathbf{x} + \mathbf{b}_1 < 0$$

$$\mathbf{W}_2^T \mathbf{W}_1^T \mathbf{x}$$

$$W_{11}^2 W_{11}^1 x + W_{21}^2 W_{12}^1 x + W_{31}^2 W_{13}^1 x + W_{41}^2 W_{14}^1 x + \mathbf{W}_2^T \mathbf{b}_1$$

$$W_{11}^2 b_1^1 + W_{21}^2 b_2^1 + W_{31}^2 b_3^1 + W_{41}^2 b_4^1 + b_1^2$$

$$x > -\frac{b_4^1}{W_{14}^1}$$

All $(W_{11}^2 W_{11}^1 + W_{21}^2 W_{12}^1 + W_{31}^2 W_{13}^1 + W_{41}^2 W_{14}^1)x + W_{11}^2 b_1^1 + W_{21}^2 b_2^1 + W_{31}^2 b_3^1 + W_{41}^2 b_4^1 + b_1^2$

$$-\frac{b_4^1}{W_{14}^1} > x > -\frac{b_3^1}{W_{13}^1}$$

Drop One $(W_{11}^2 W_{11}^1 + W_{21}^2 W_{12}^1 + W_{31}^2 W_{13}^1)x + W_{11}^2 b_1^1 + W_{21}^2 b_2^1 + W_{31}^2 b_3^1 + b_1^2$

$$-\frac{b_3^1}{W_{13}^1} > x > -\frac{b_2^1}{W_{12}^1}$$

Drop Two $(W_{11}^2 W_{11}^1 + W_{21}^2 W_{12}^1)x + W_{11}^2 b_1^1 + W_{21}^2 b_2^1 + b_1^2$

$$-\frac{b_2^1}{W_{12}^1} > x > -\frac{b_1^1}{W_{11}^1}$$

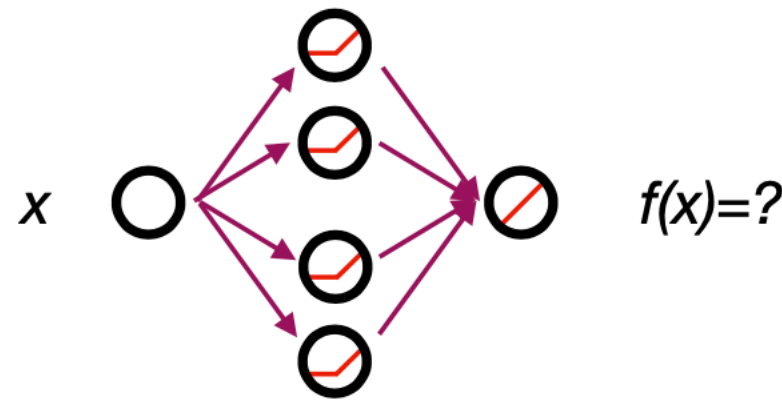
Drop Three $(W_{11}^2 W_{11}^1)x + W_{11}^2 b_1^1 + b_1^2$

$$x < -\frac{b_1^1}{W_{11}^1}$$

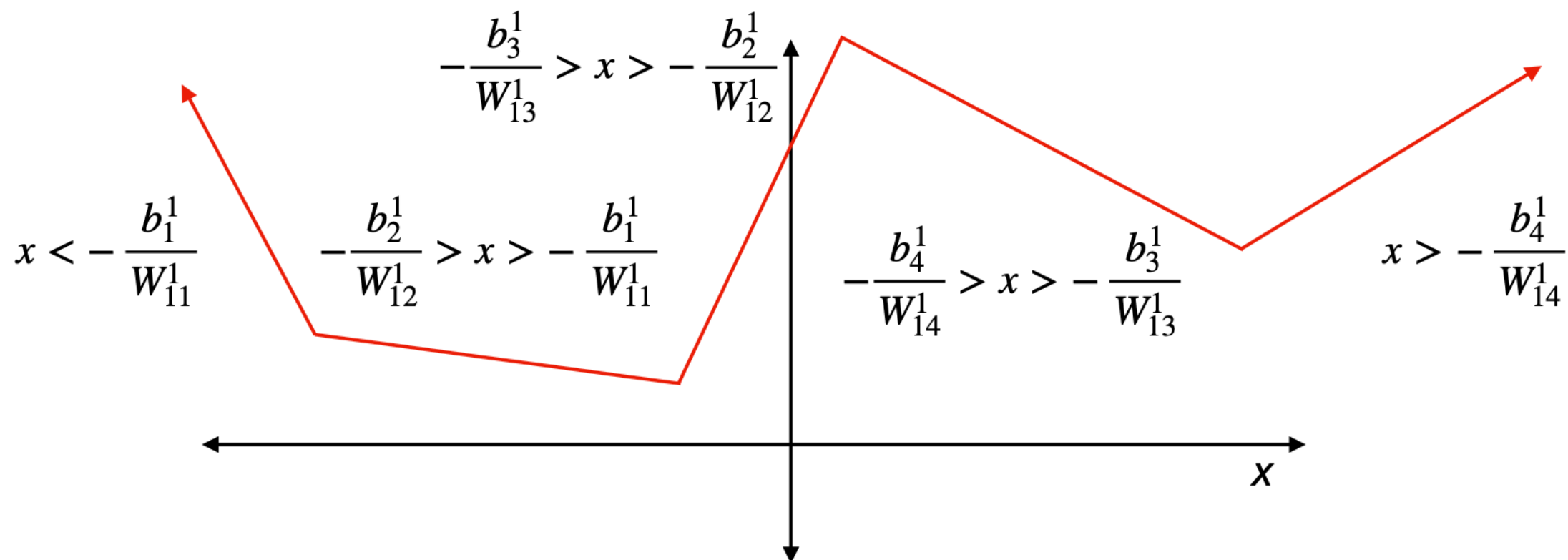
Drop It b_1^2 **Like it's Hot**

Universal Functions

Approximating (Example)

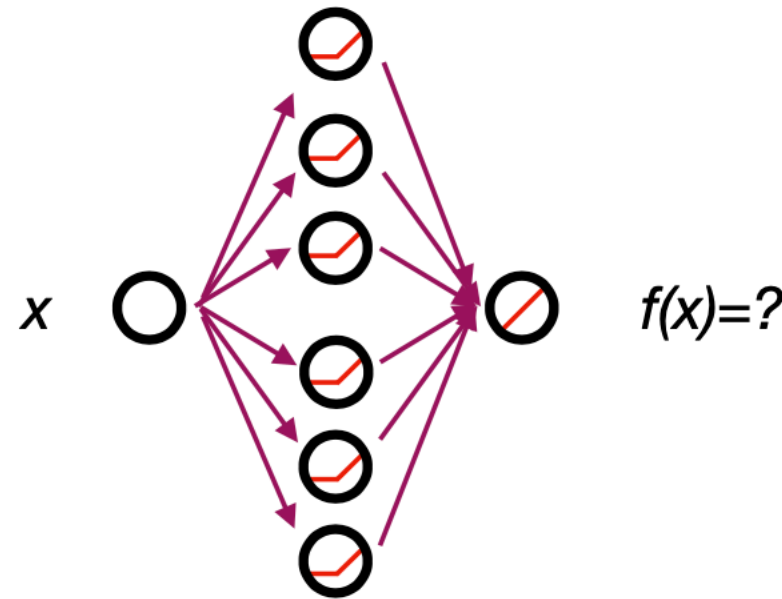


What kind of function can this network architecture approximate?

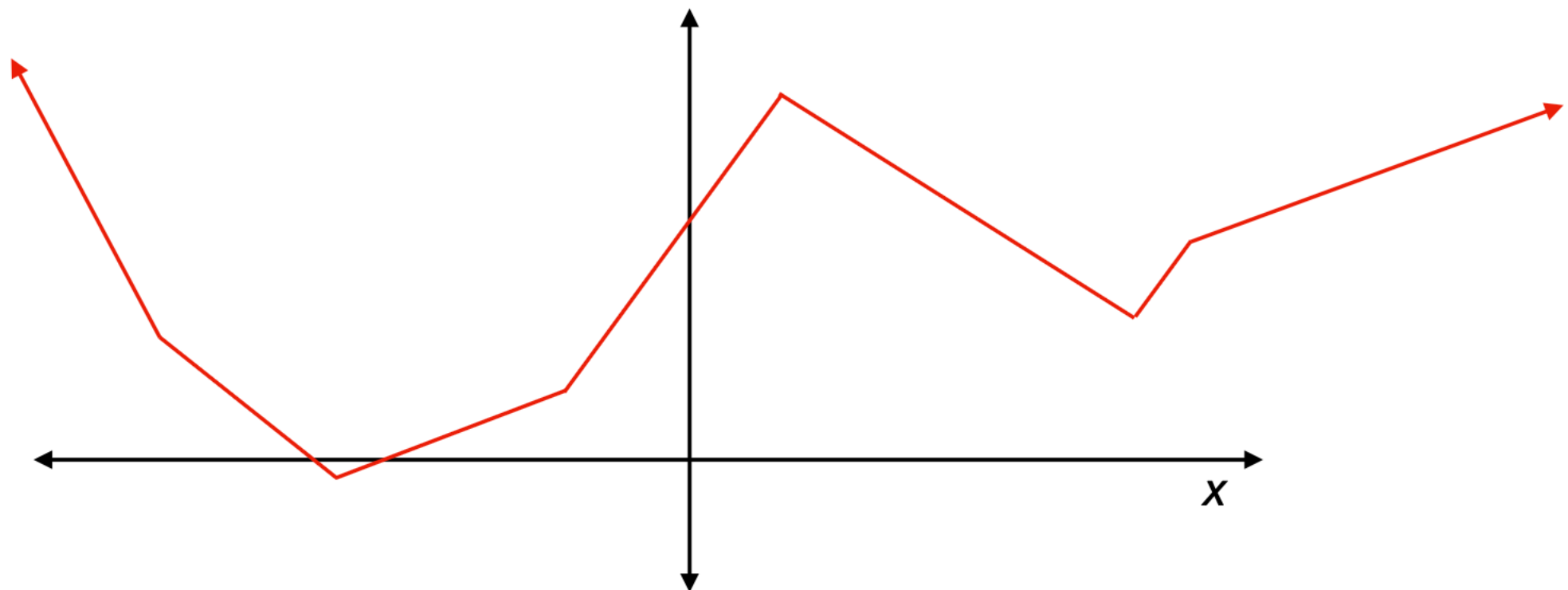


Universal Functions

Approximating (Example)

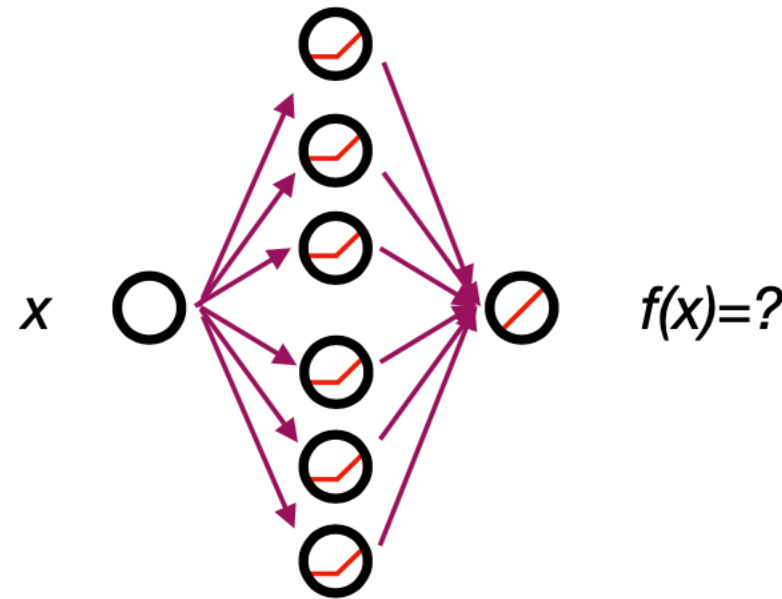


What kind of function can this network architecture approximate?

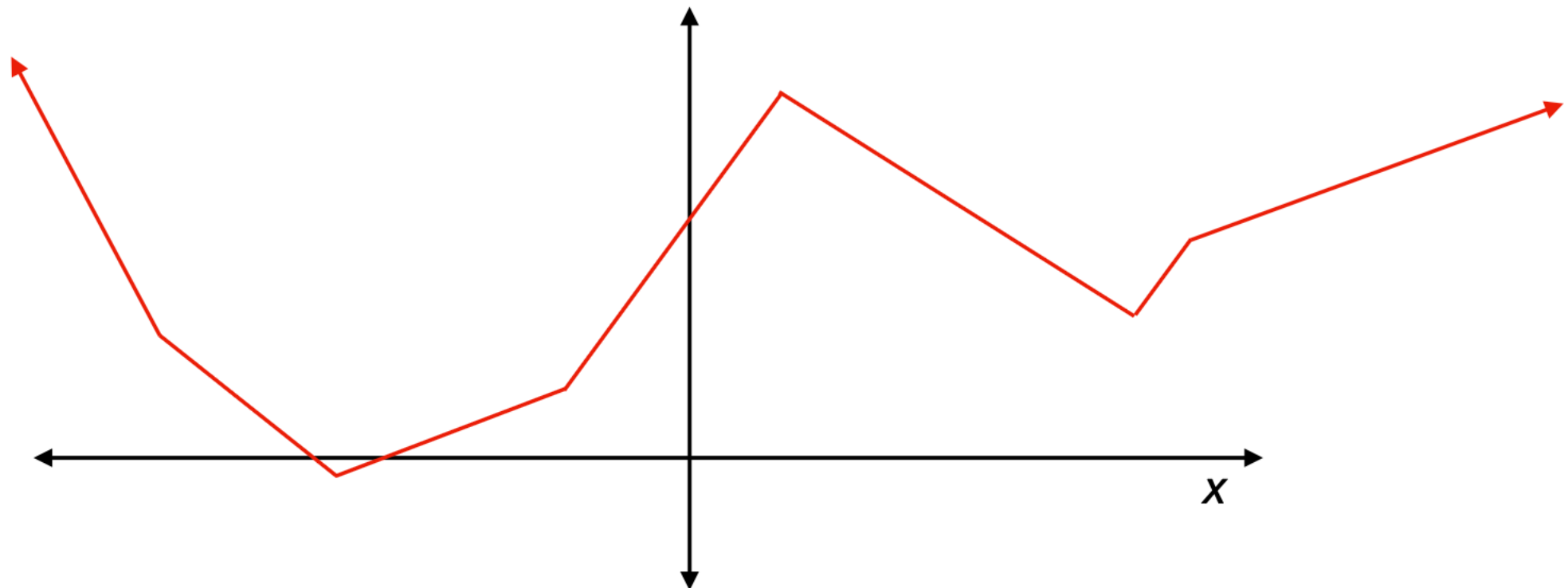


Universal Functions

Approximating (Example)



What kind of function can this network architecture approximate?



Lets Solve A Real Problem

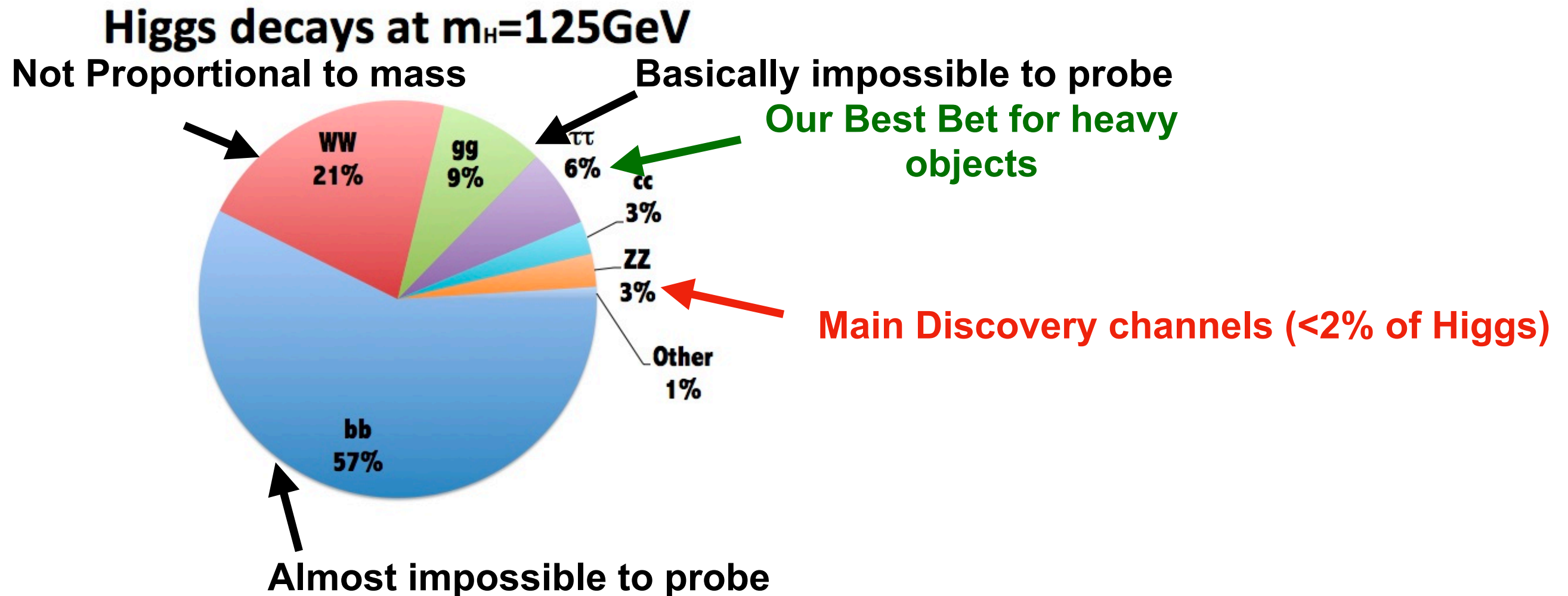
- Let's look at the tau lepton



	<i>Quarks</i>		<i>Leptons</i>	
<i>Generation 3</i>	 t Top	 b Bottom	 τ Tau	 ν_τ Tau-neutrino
<i>Generation 2</i>	 c Charm	 s Strange	 μ Muon	 ν_μ Muon-neutrino
<i>Generation 1</i>	 u Up	 d Down	 e Electron	 ν_e Electron-neutrino

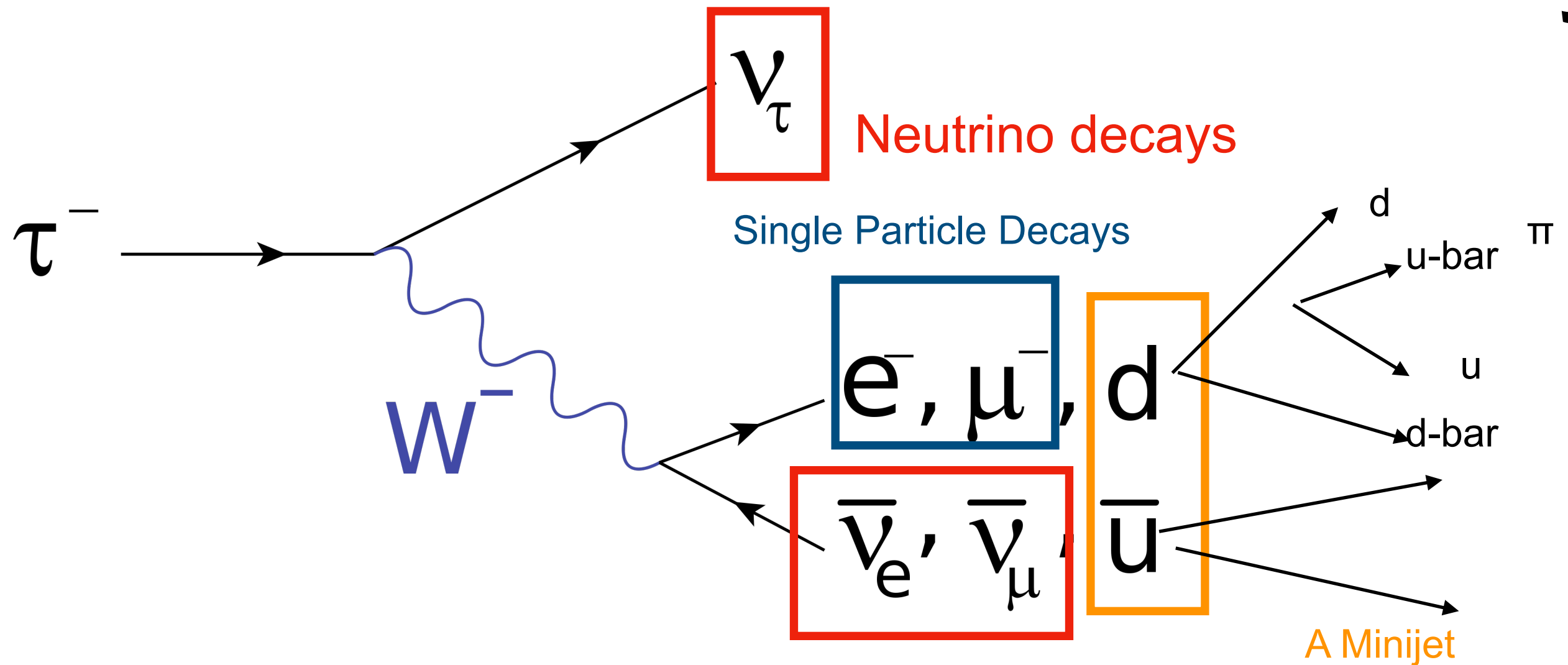
The Tau is the heaviest of the leptons (electron-like)
What makes it so special?

Higgs Decays



- Higgs probability of decay to quarks and leptons is proportional the mass of the particle. Taus are very heavy particles. Higgs decays to them 6% of the time. That's great. **It was the first channel we could actually probe the proportionality to mass.**

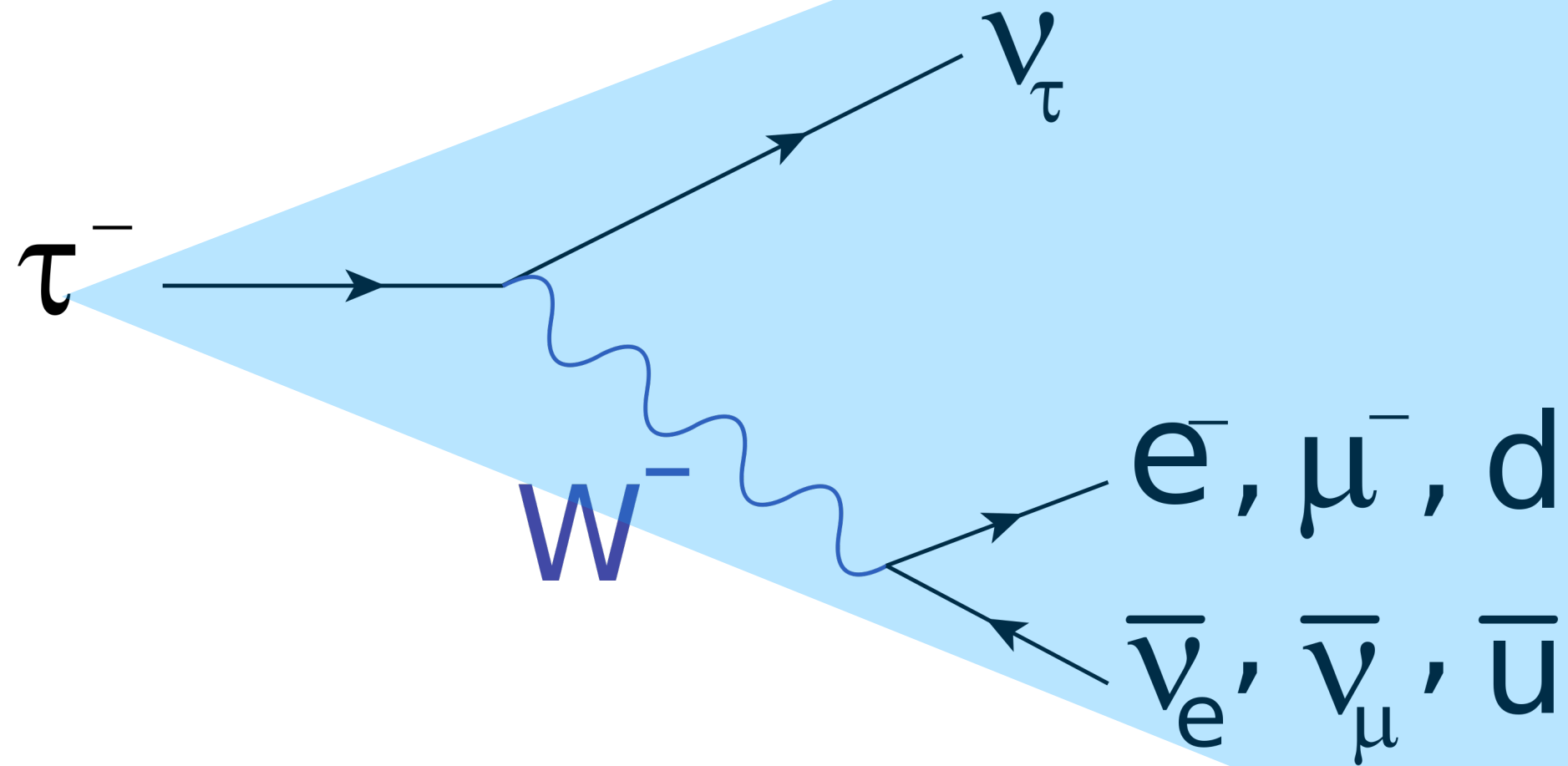
Tau Decays



Neutrino Decays: The probability of a neutrino interaction is too small to see at the LHC. These particles are invisible

Single Particle decays: These events just give us one particle e or μ
 Minijet: Decays to quarks give us a shower of particles in small jet

Problem

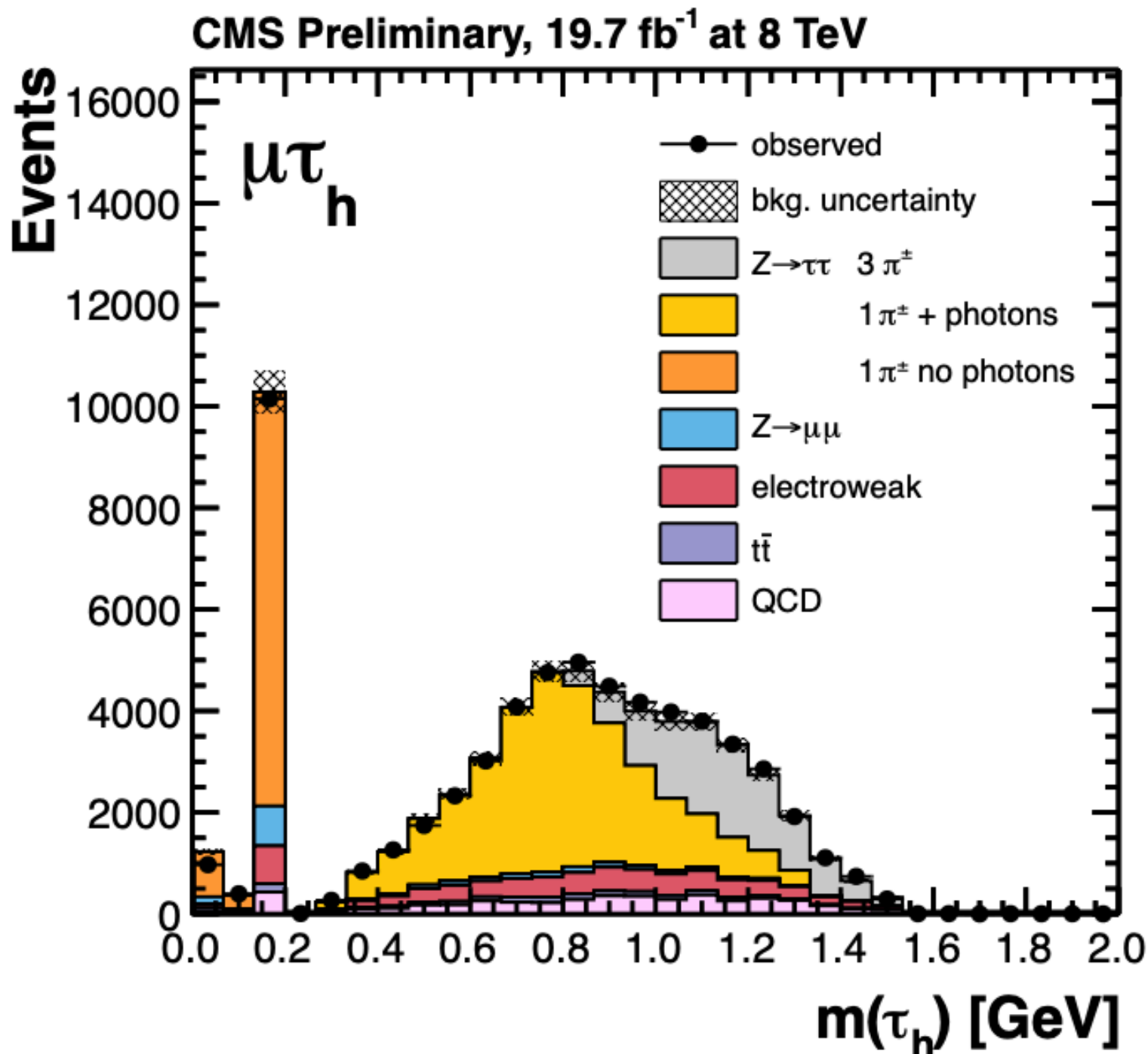


Take a jet
And Sum all the particles

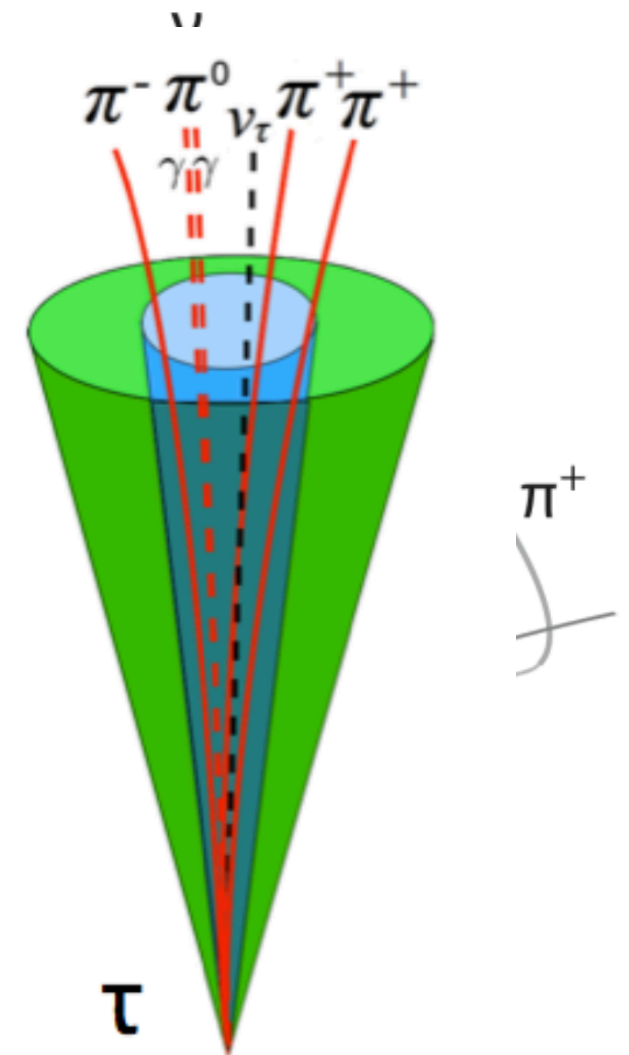
Can we go from
Jet $p \rightarrow$ Tau p

Can we guess direction of the neutrinos and reconstruct the original tau energy?

How does a Tau decay

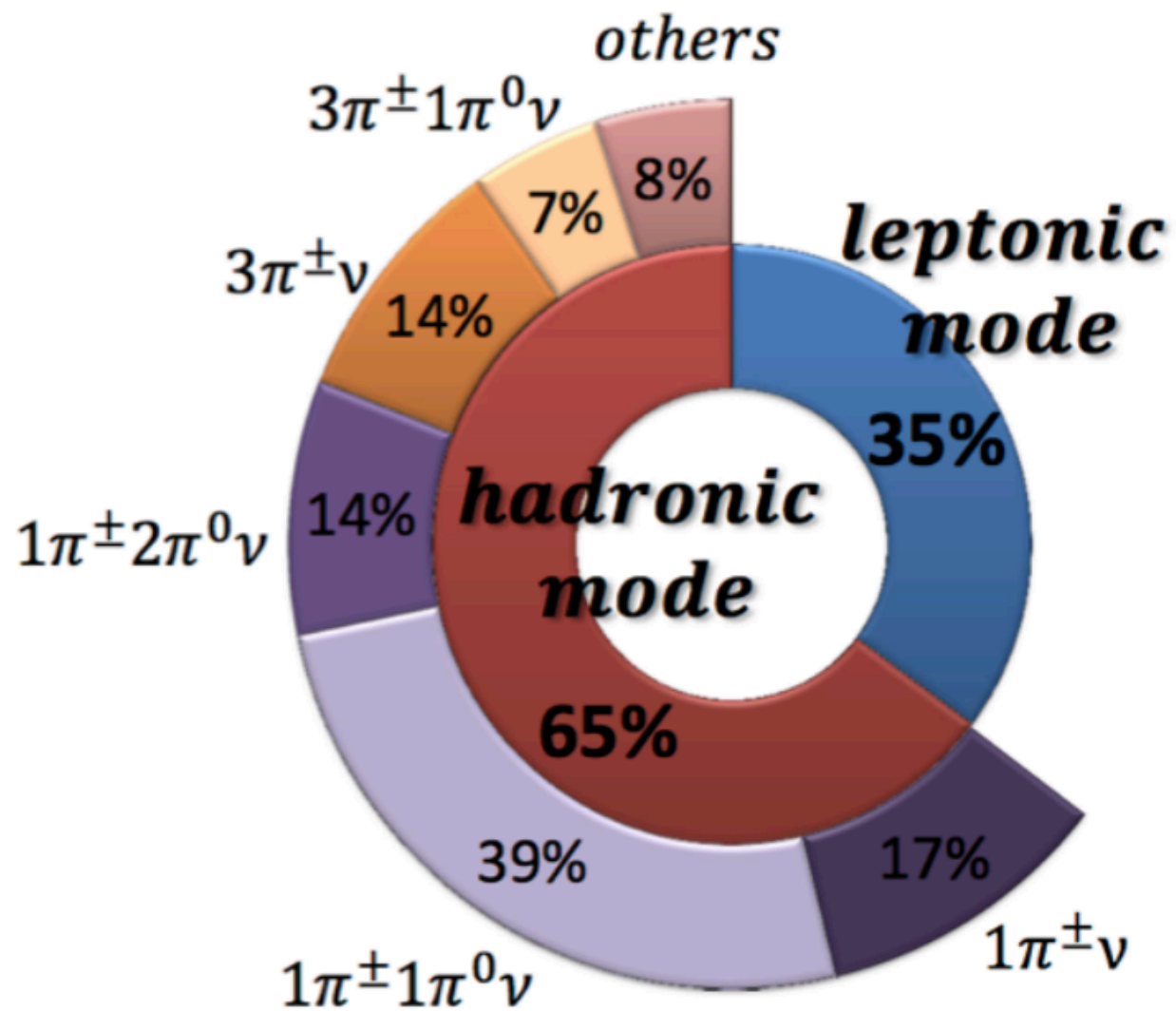


$$m_\tau = 1.76 \text{ MeV}$$

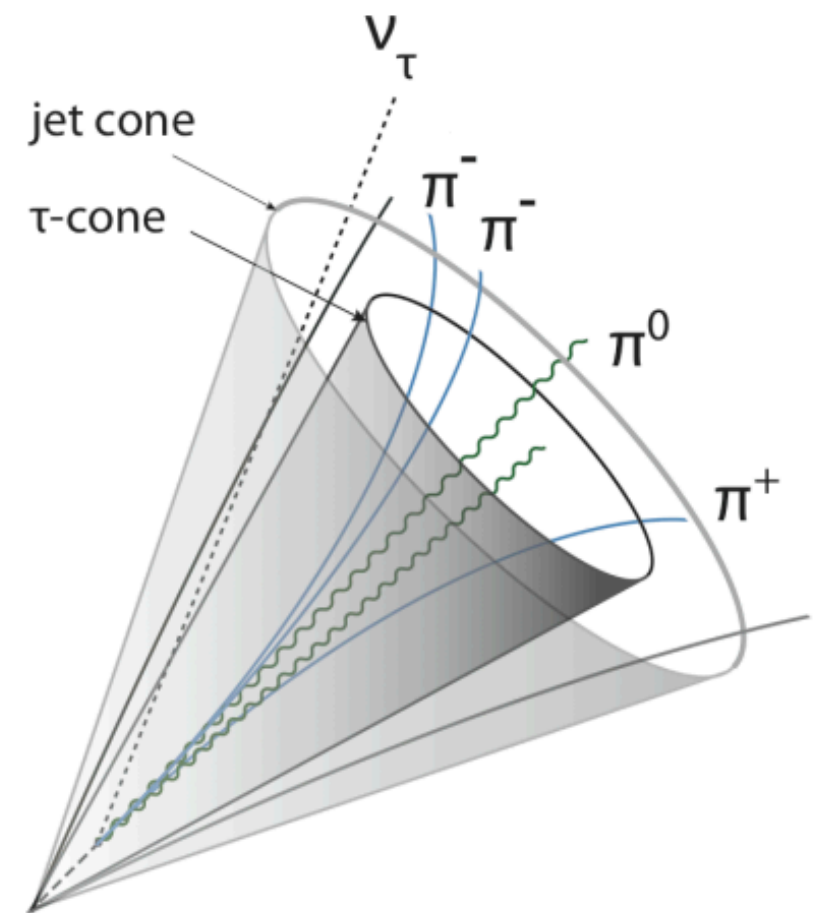


Taus have a small mass, which means they can be found within a small cone

How does a Tau decay

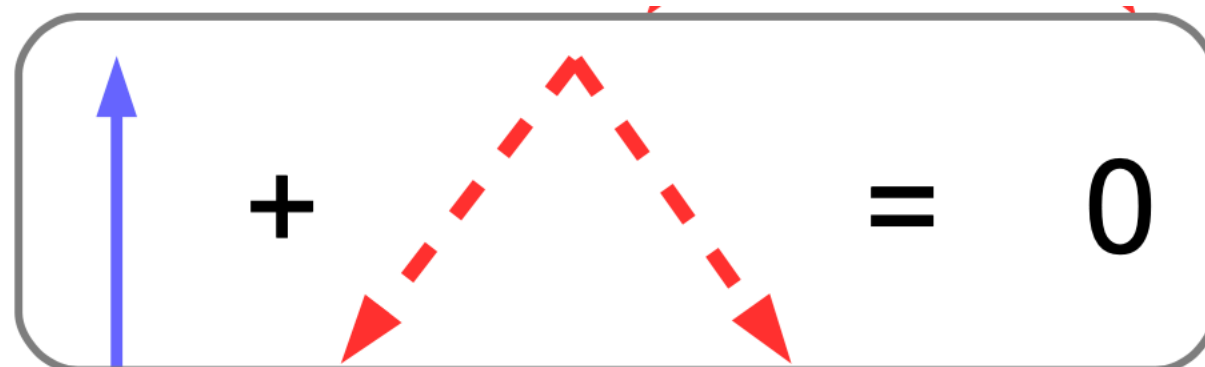
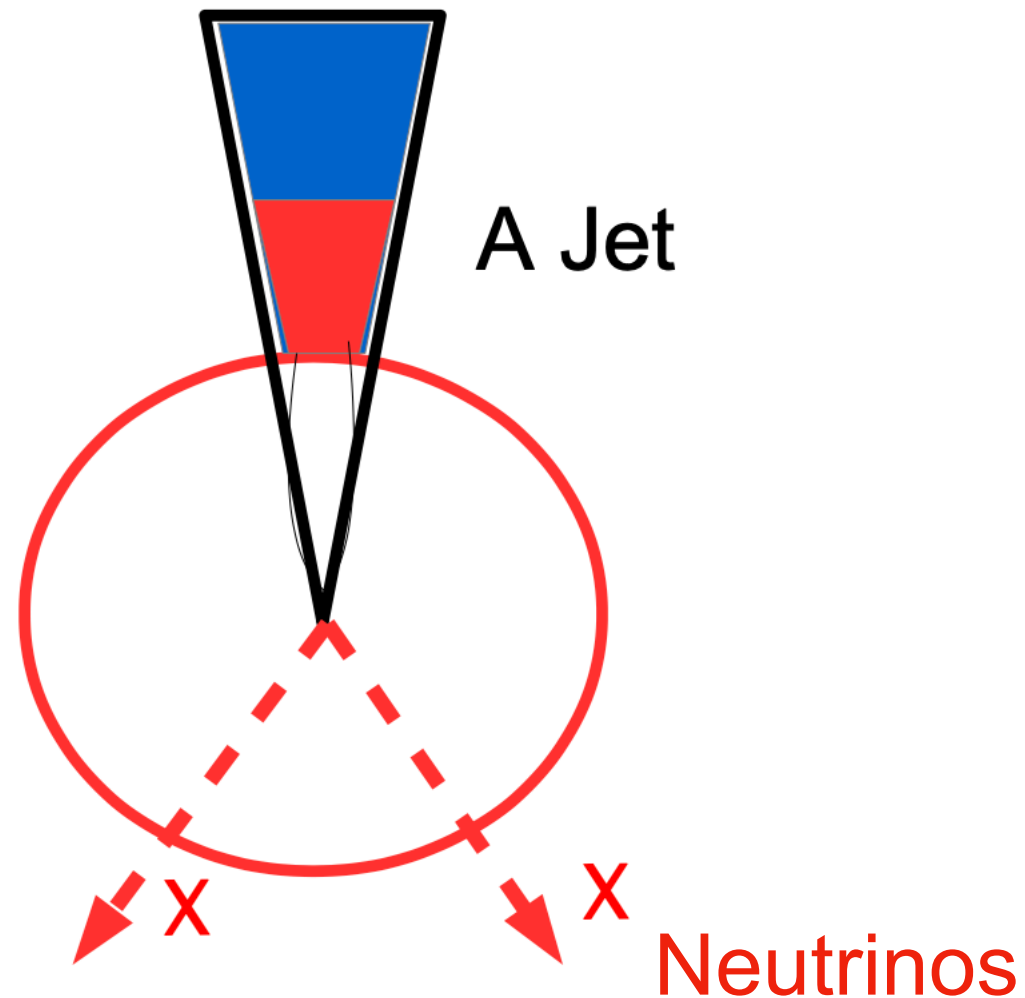


$$\pi^0 \rightarrow \gamma\gamma$$



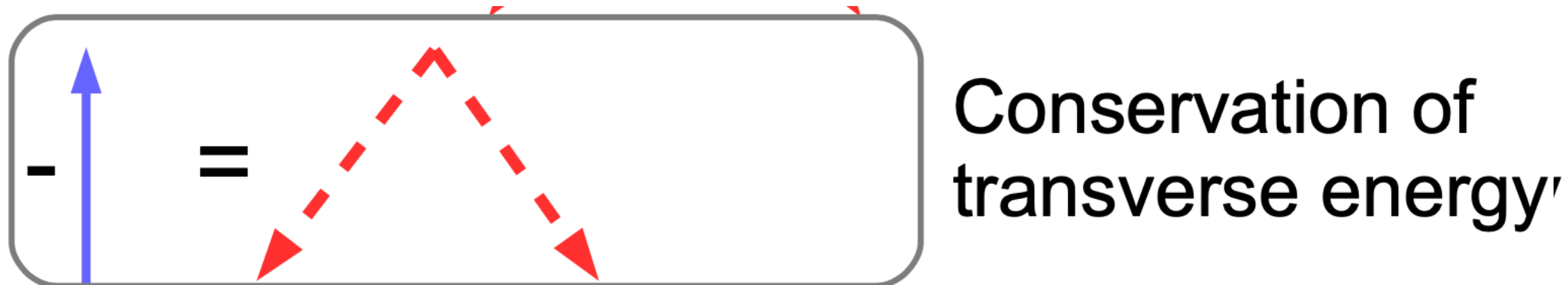
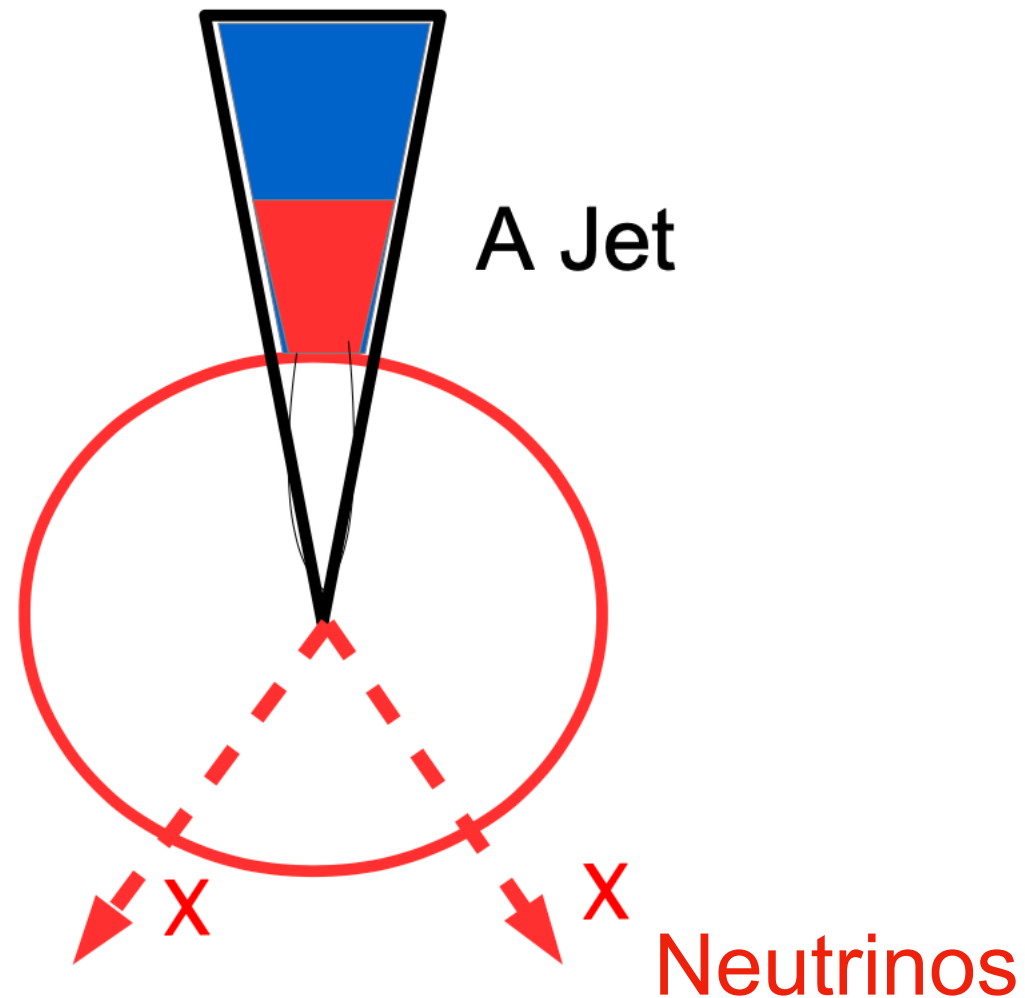
We are looking for collection of 1-5 particles
Neutrino will fall in the same cone

What we did for that result

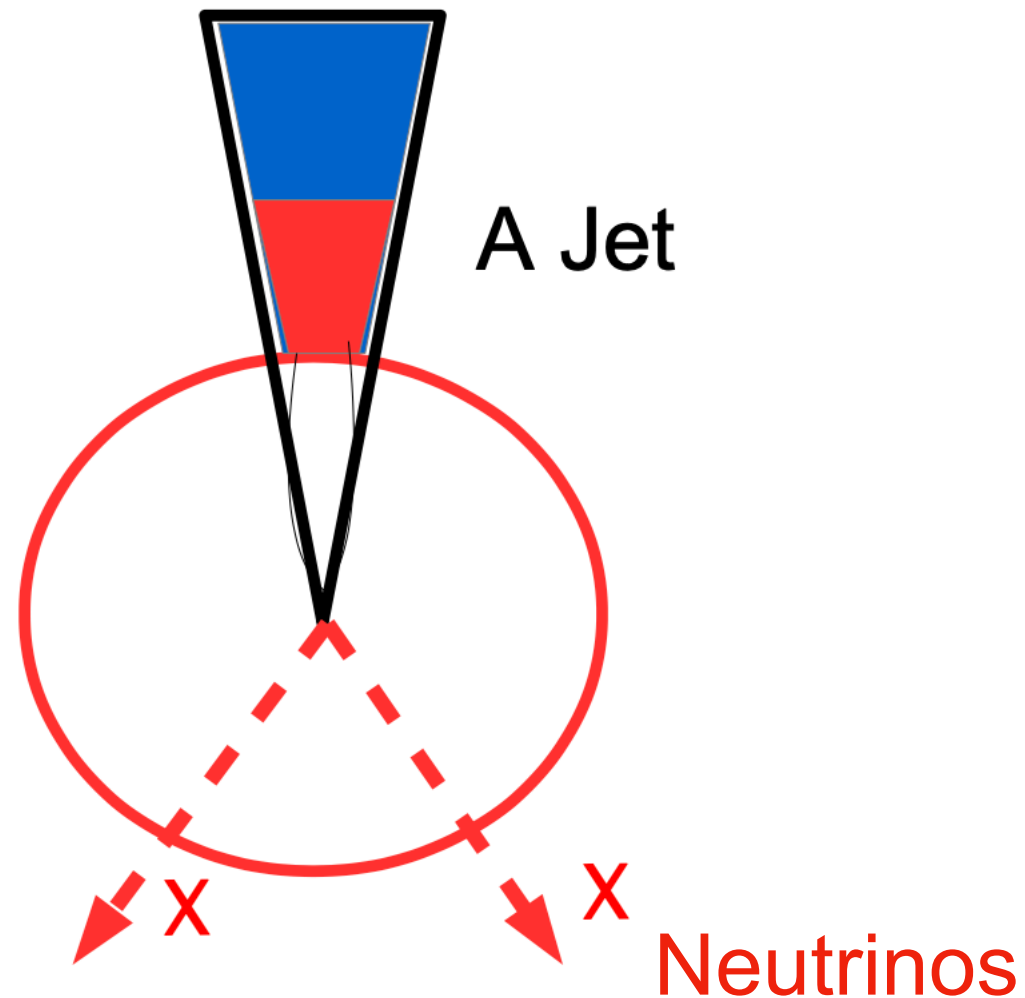


Conservation of
transverse energy

What we did for that result



What we did for that result

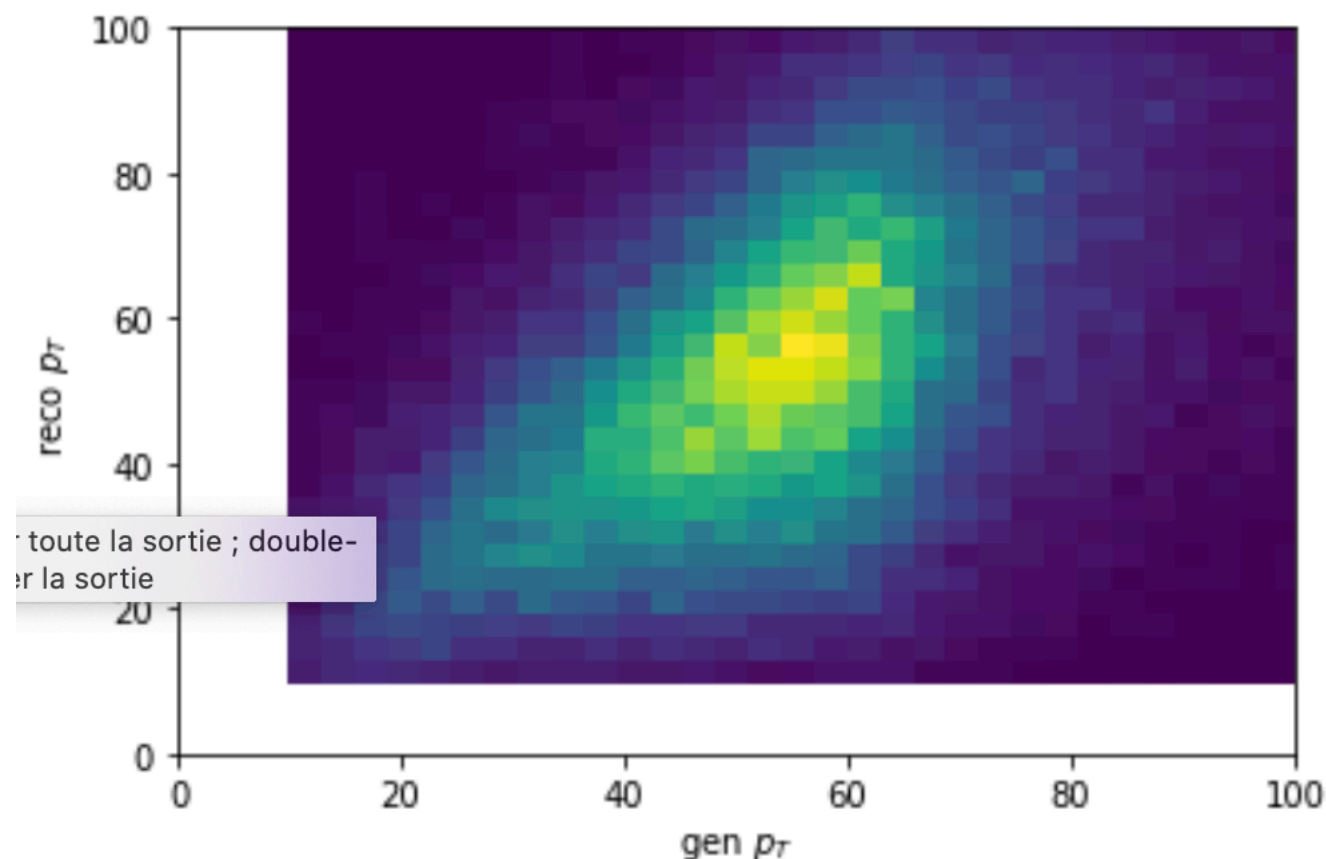


$$-\sum_{\text{All particles}} \vec{p}_T = \overrightarrow{MET} \quad (E_T^{\text{Miss}})$$

Conservation of
transverse energy

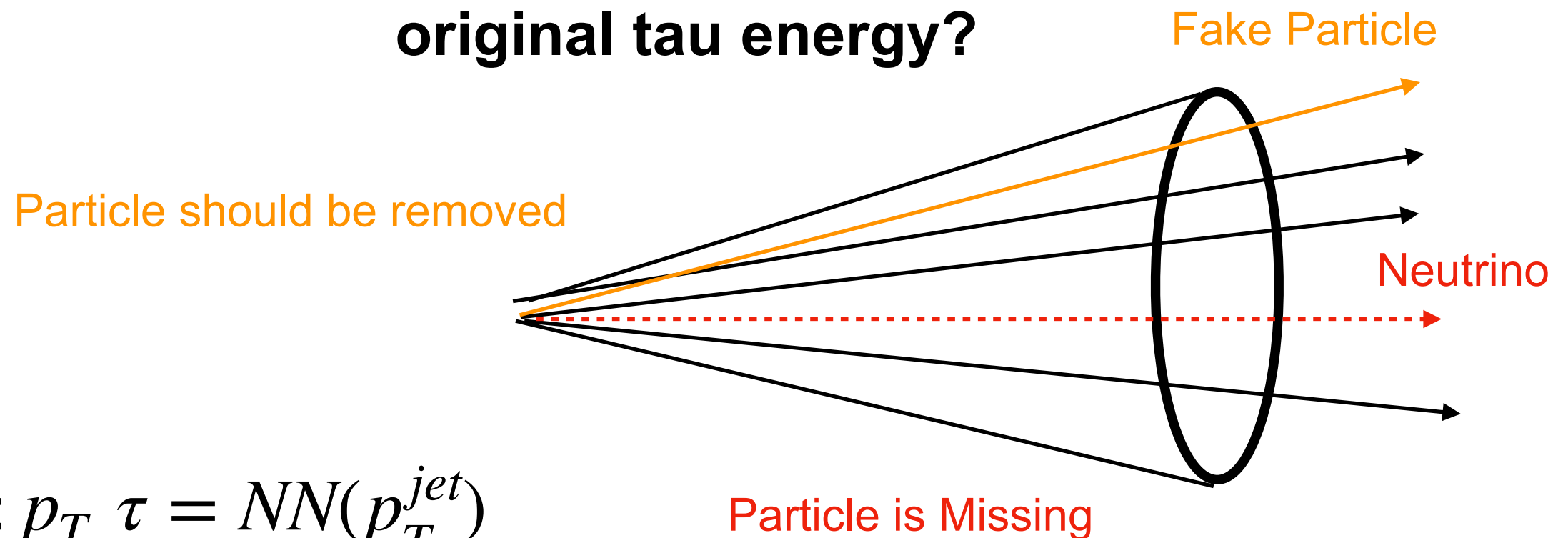
Some Correlation

- In this case, we want to try to use the tau momentum
- Goal here is to rely on the fact that there is some correlation
- The tau momentum can predict the total tau energy



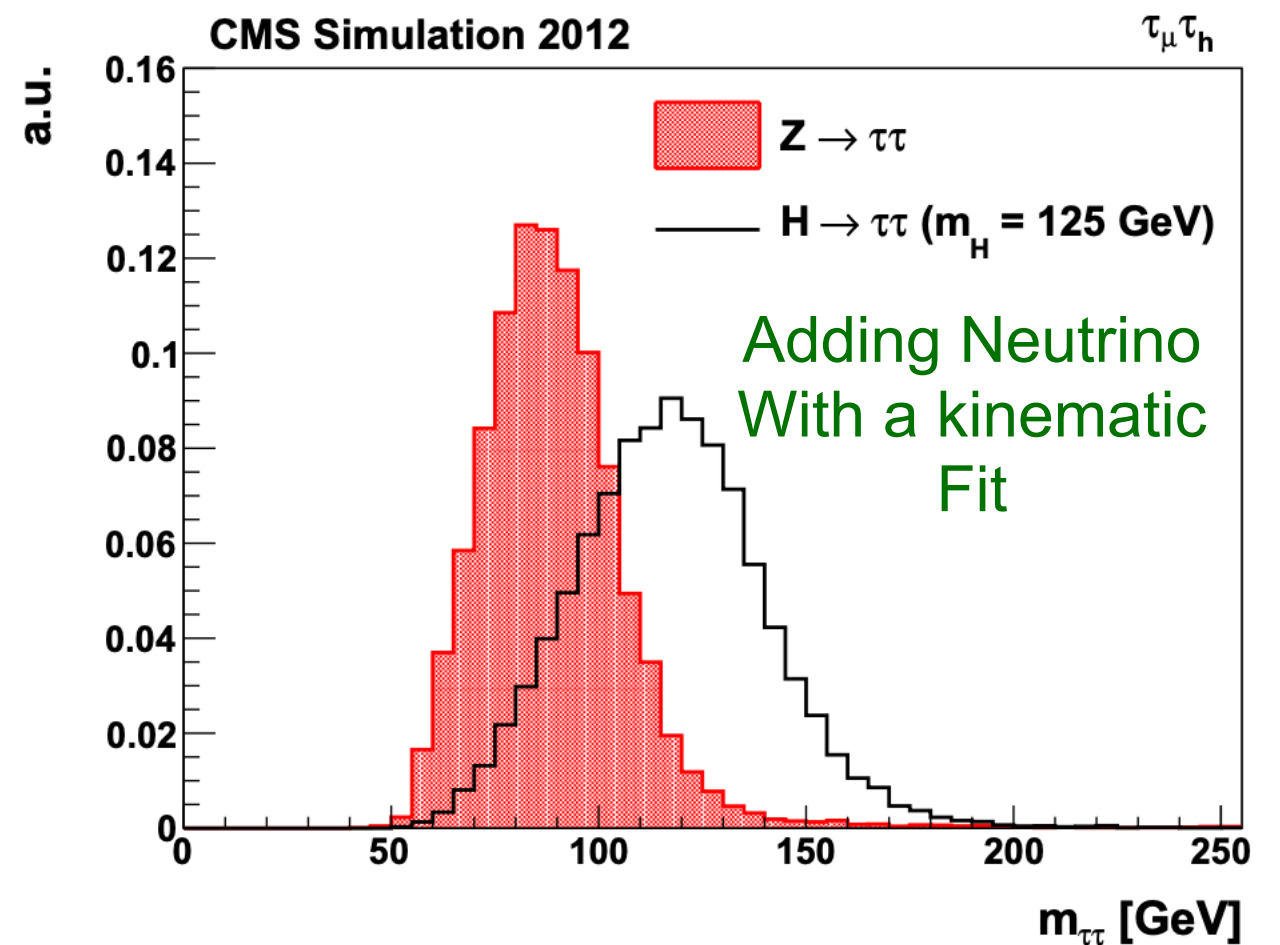
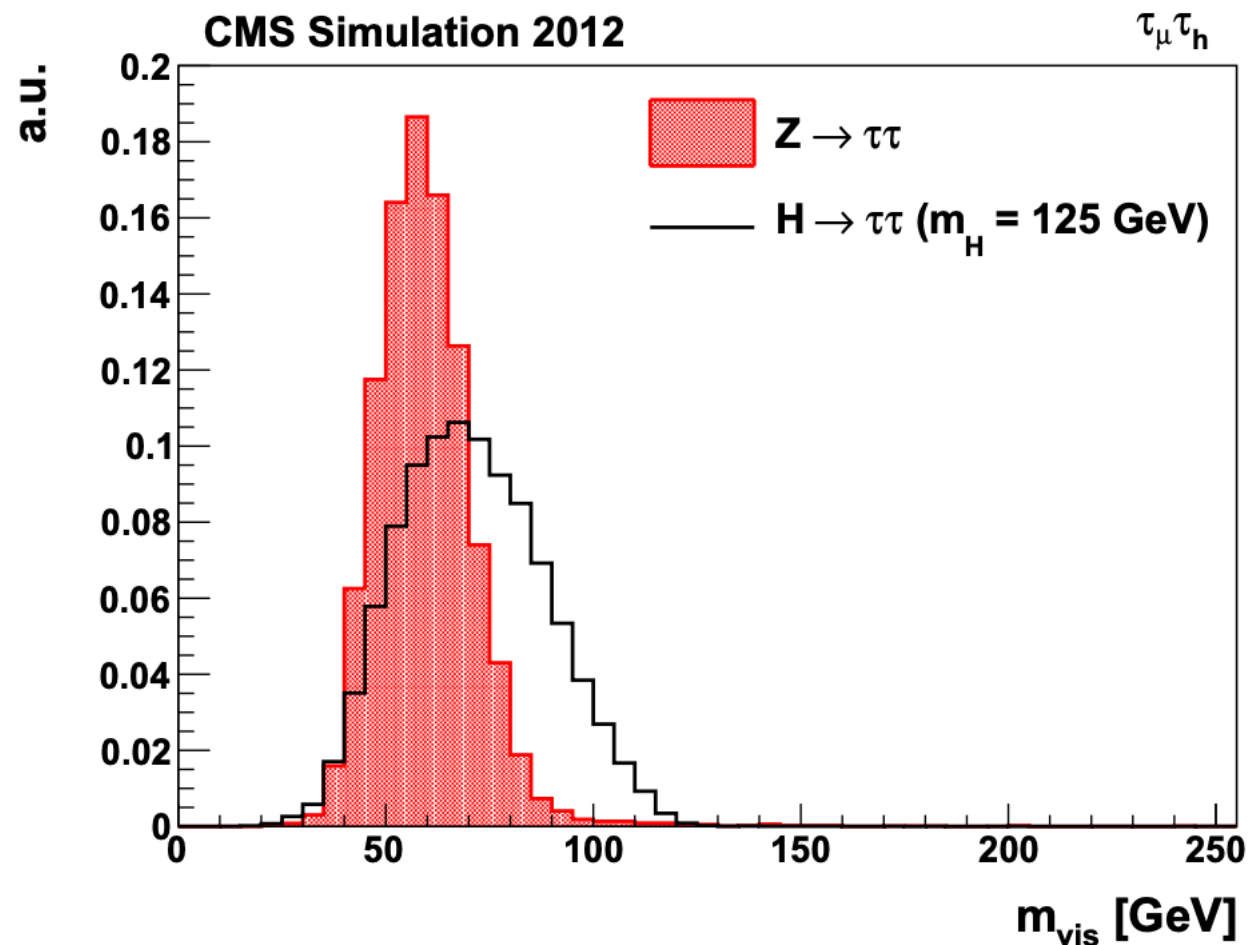
NN Problem

Can we guess direction of the neutrinos and reconstruct the original tau energy?



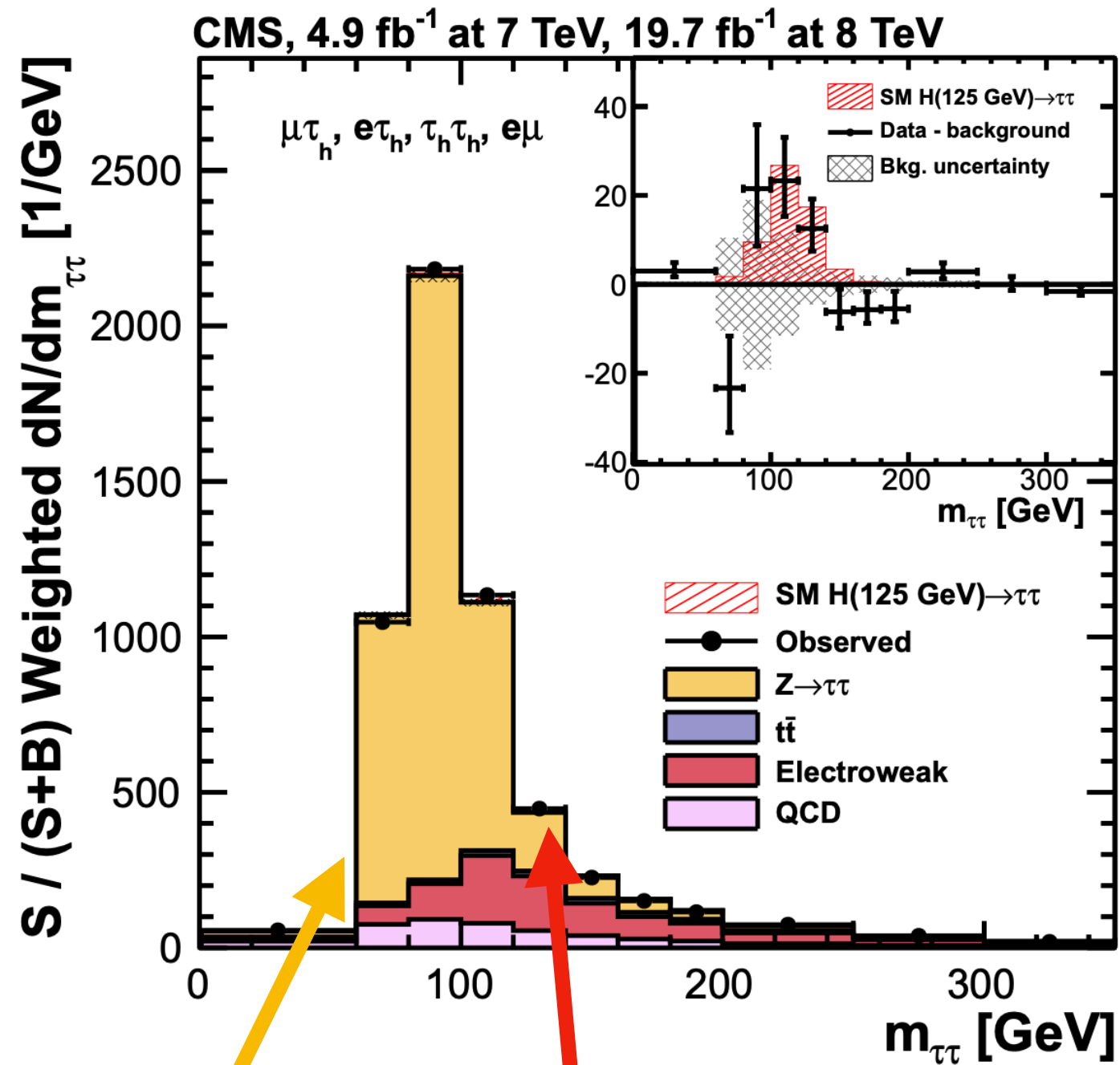
- simple: $p_T \tau = NN(p_T^{jet})$
- reduced scale: $\frac{p_T \tau}{p_T^{jet}} = NN(p_T^{jet})$
- Complex: $\frac{p_T \tau}{p_T^{jet}} = NN(\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4, \vec{p}_5)$

Why this?



- Finding the Higgs boson is hard we need to separate
 - Higgs boson mass peak from the **Z boson mass**
- When Higgs discovered didn't have the NN tech to add neutrinos

The Full Challenge



Background

Signal

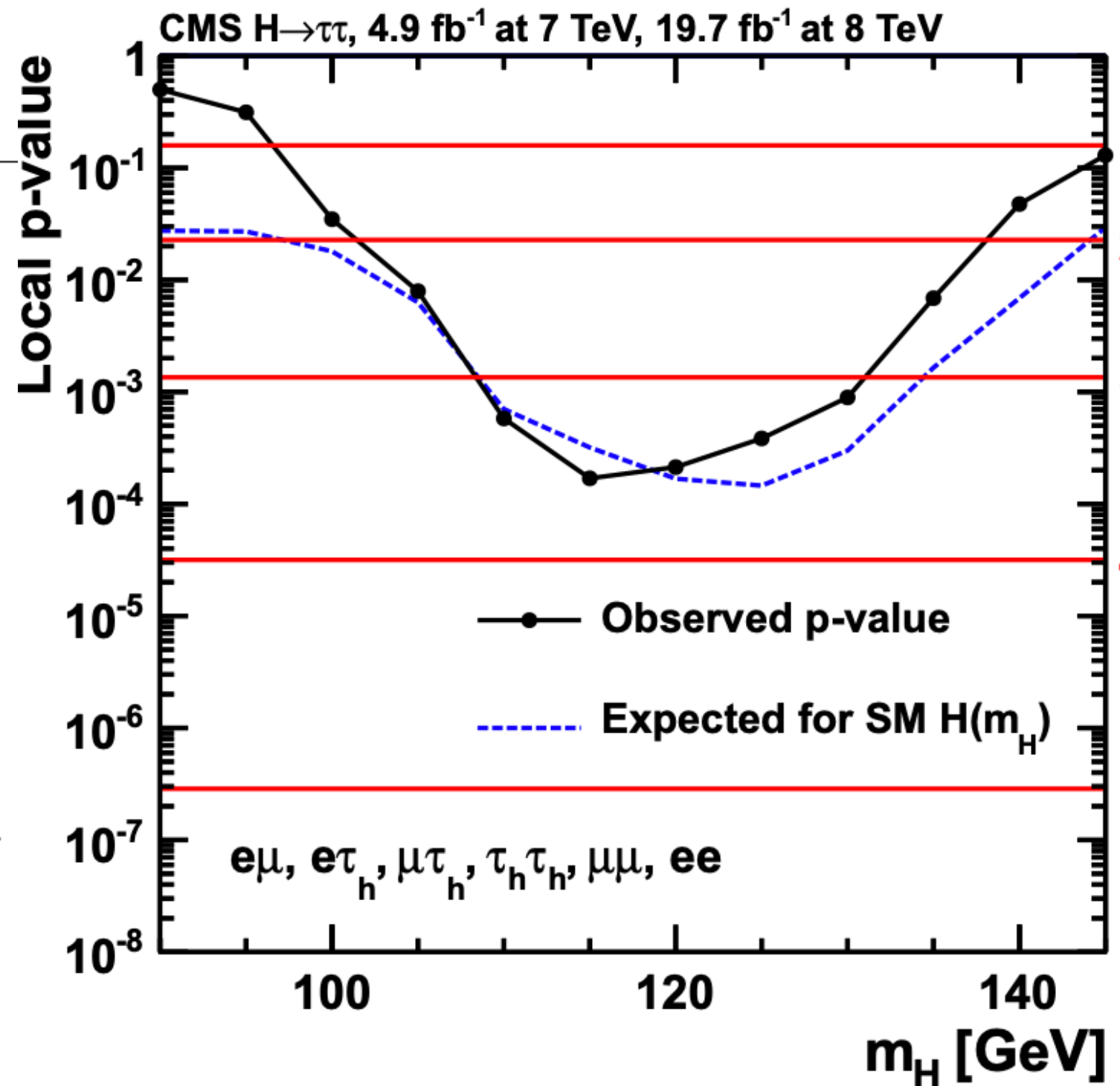
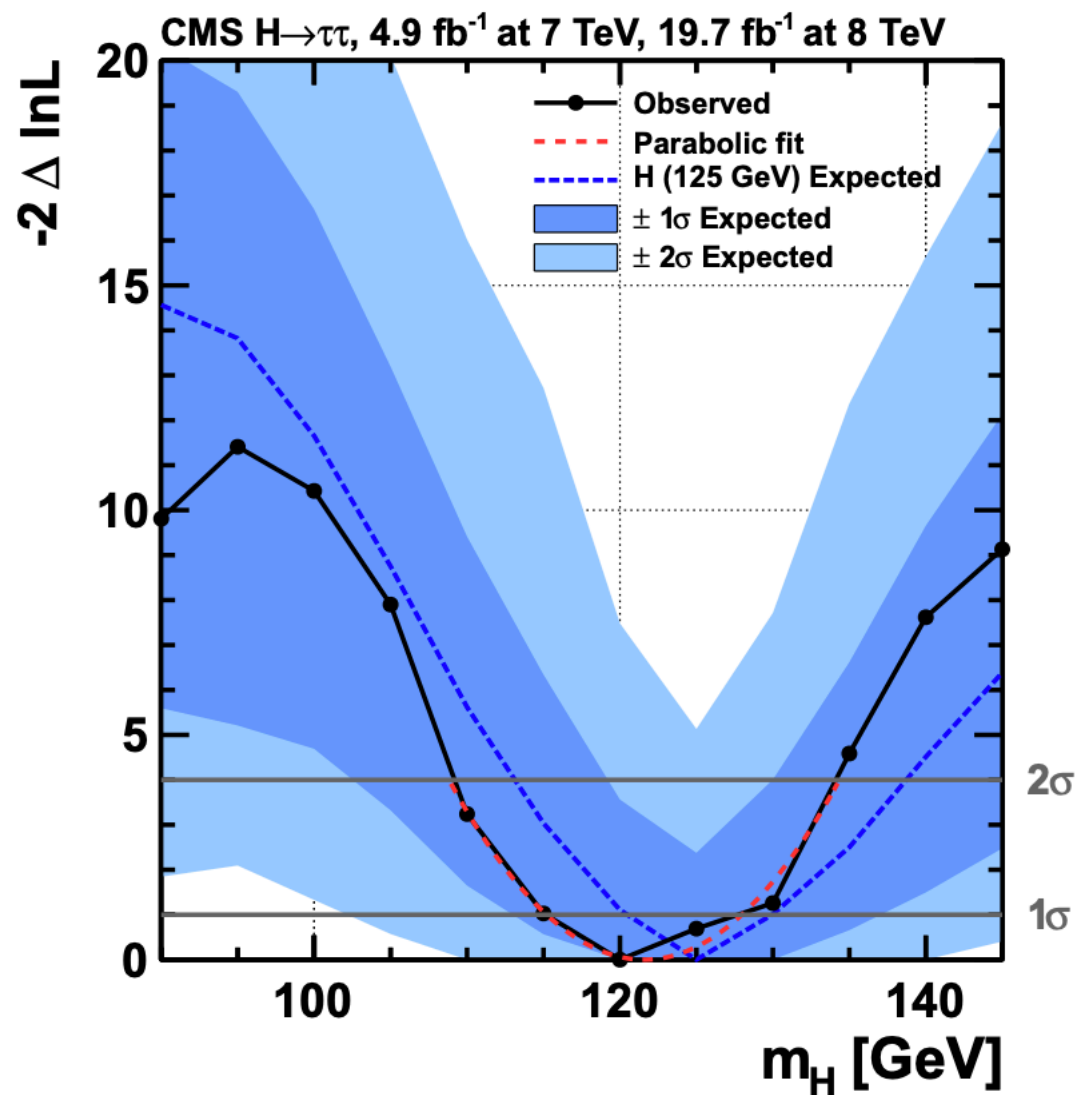
Plot is a composite
of 70 separate fits

There were > 2000
Floated parameters

Fit took 24h to run

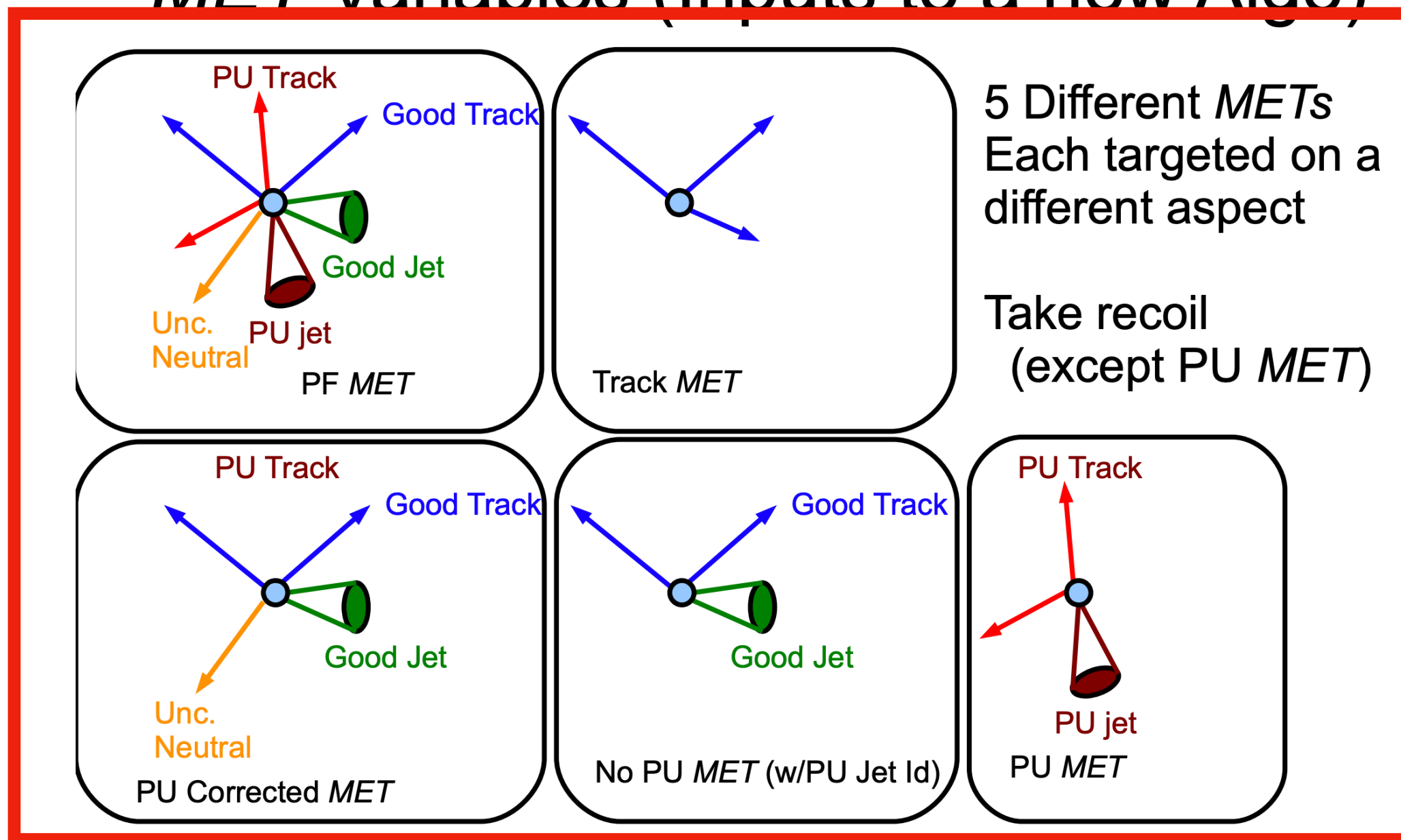
Higgs to Tau Tau Bound

- Best fit



What we did for that result

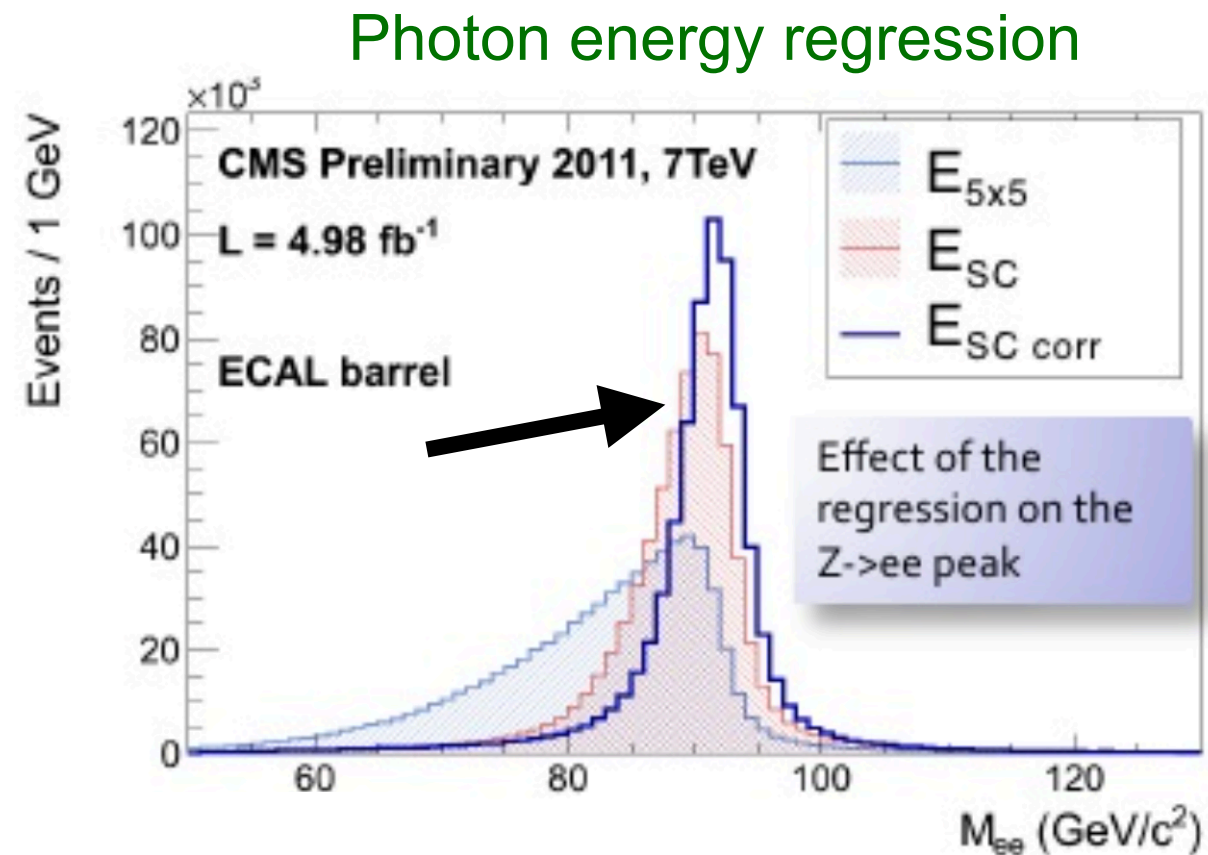
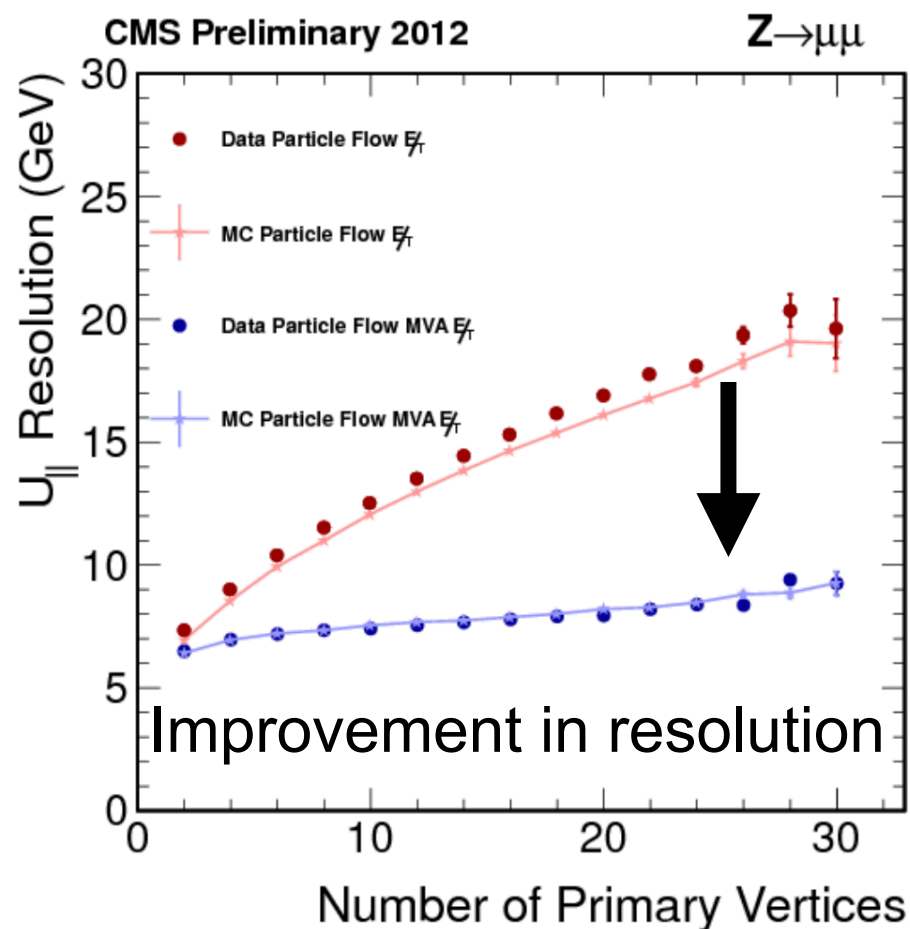
MET Variables (Inputs to a new Algo)



All of these separate *MET* calculations were put into 1 single regression

- We did end up a using an NN regression for that plot

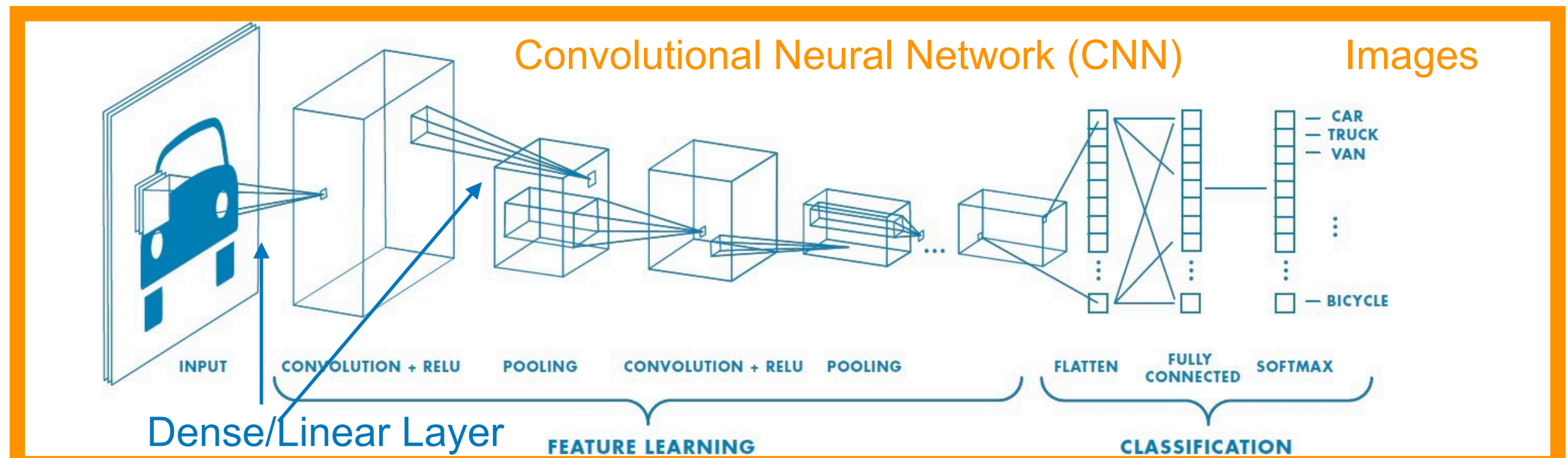
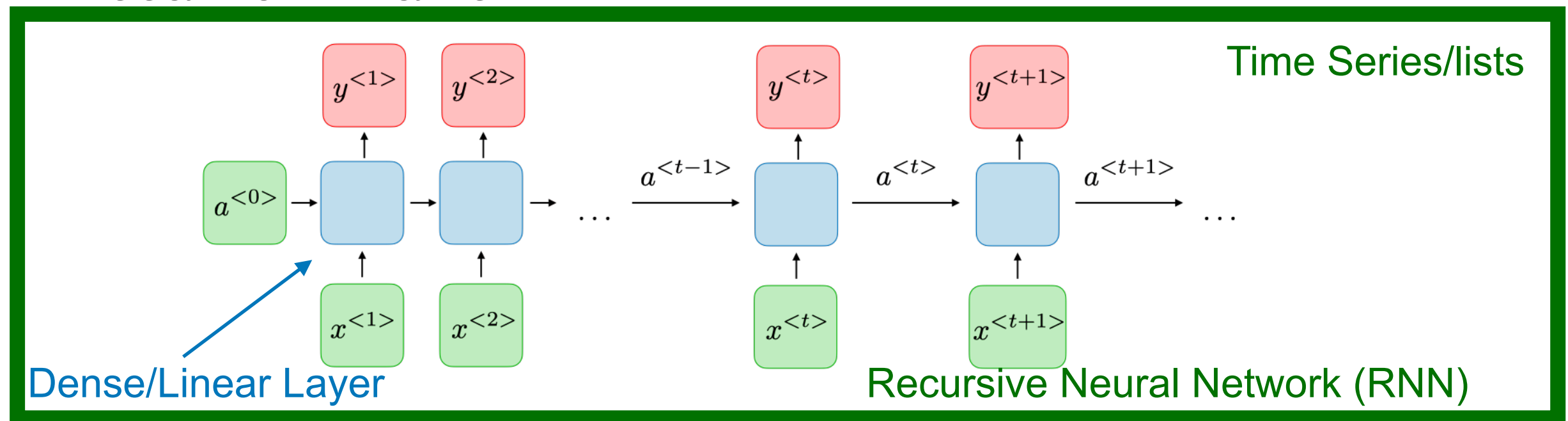
Impact of Regression



- Regression ended improving the Higgs sensitivity by 30%
 - Both in the diphoton channel and Higgs to tau leptons
 - This is teh difference between 2σ and 3σ

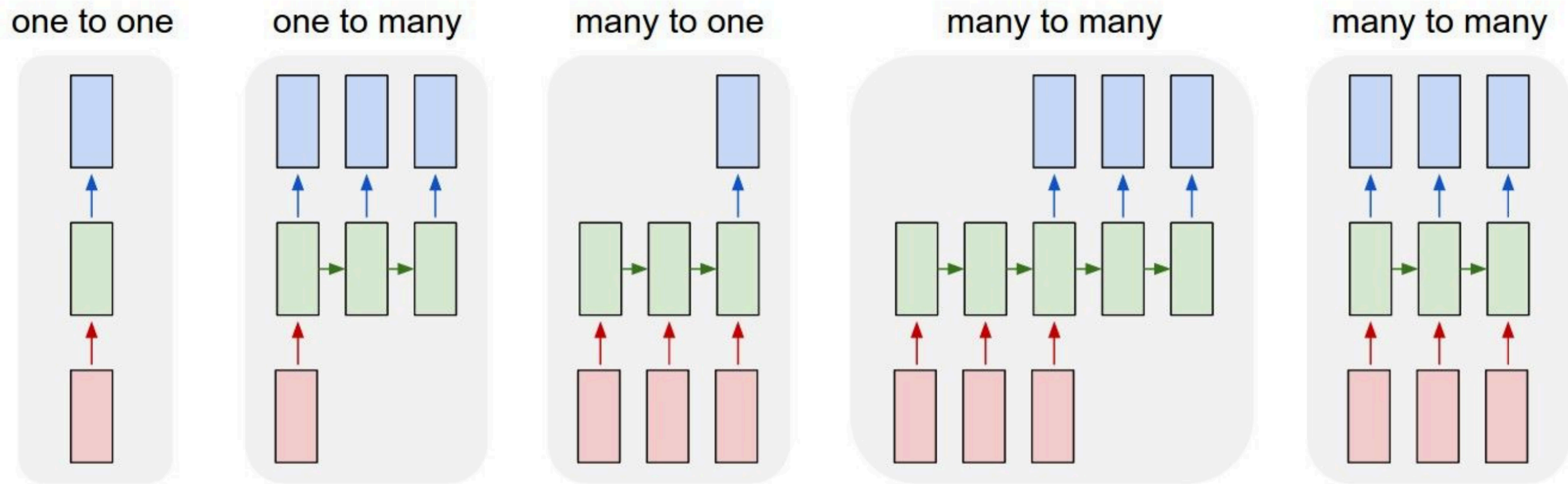
When are different NN³⁵ geometries useful?

- Recall from Earlier



Using an RNN

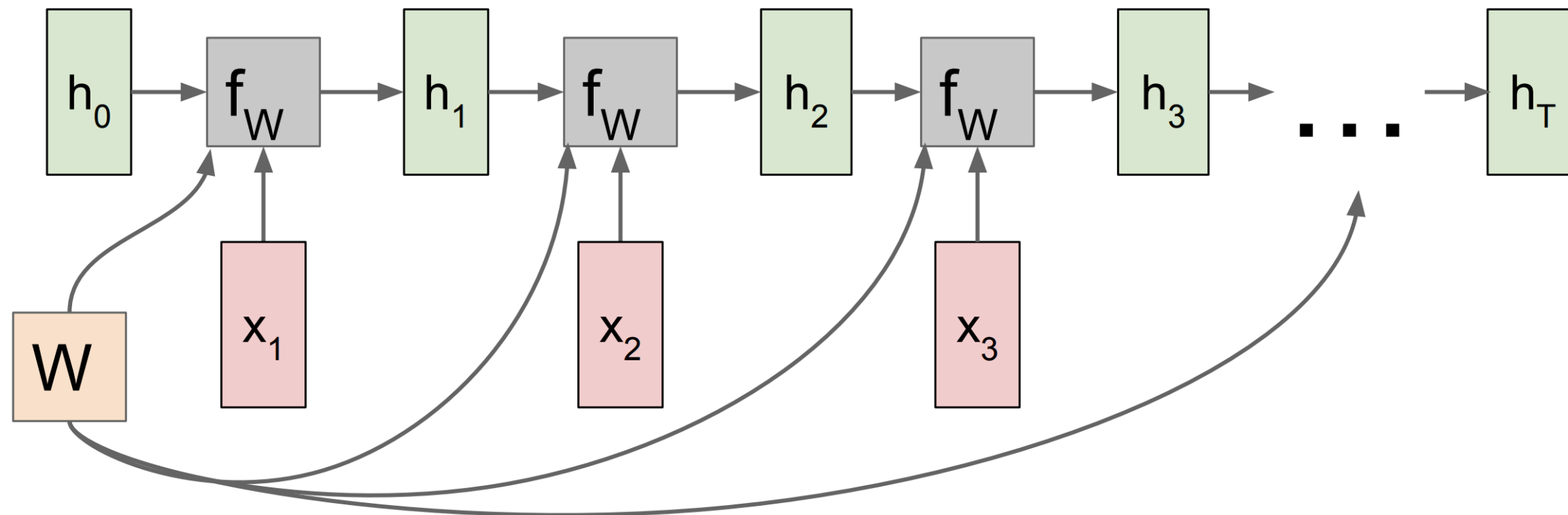
- Recursive neural network takes input one by tone



↖ e.g. **Sentiment Classification**
sequence of words -> sentiment

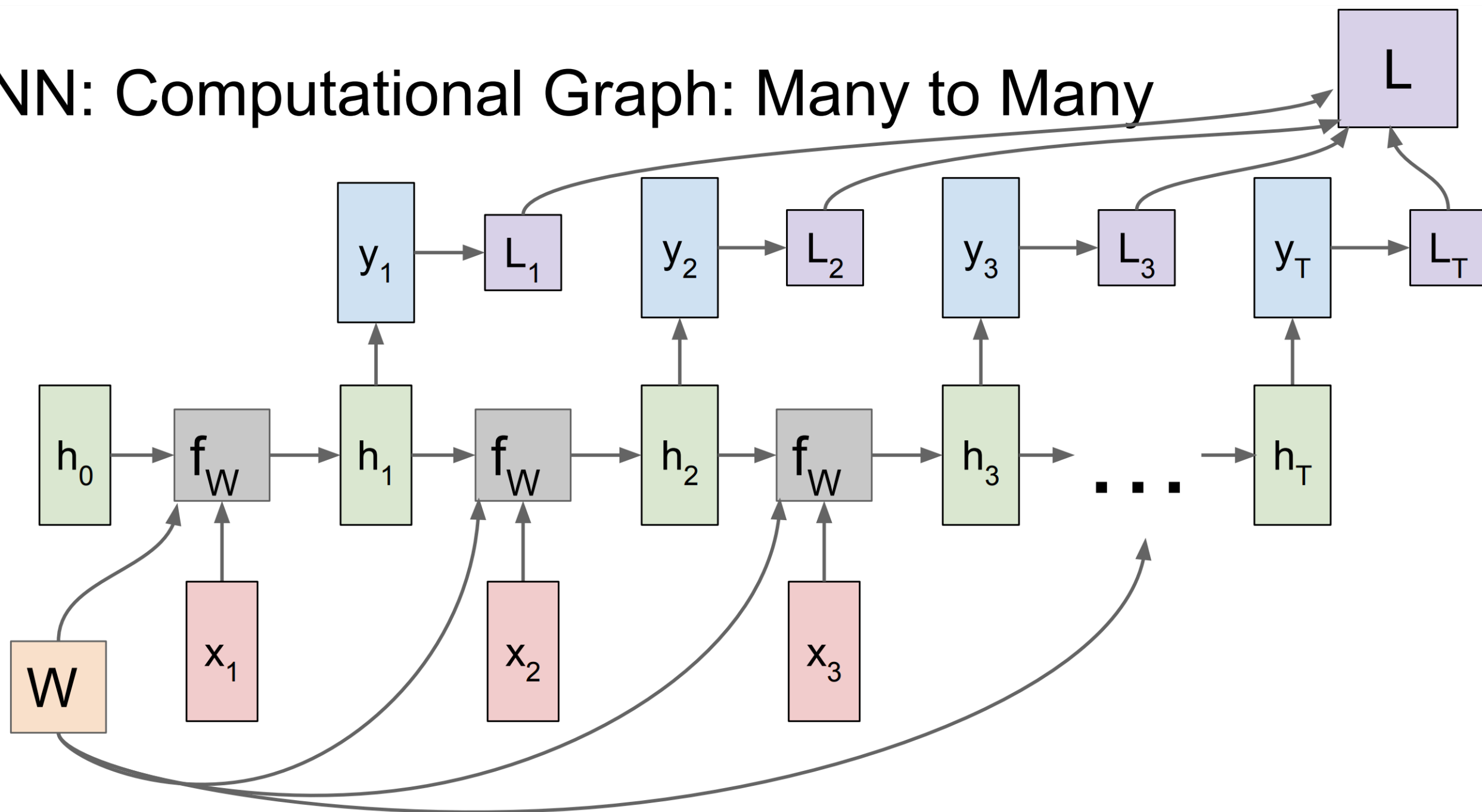
Using an RNN

Re-use the same weight matrix at every time-step

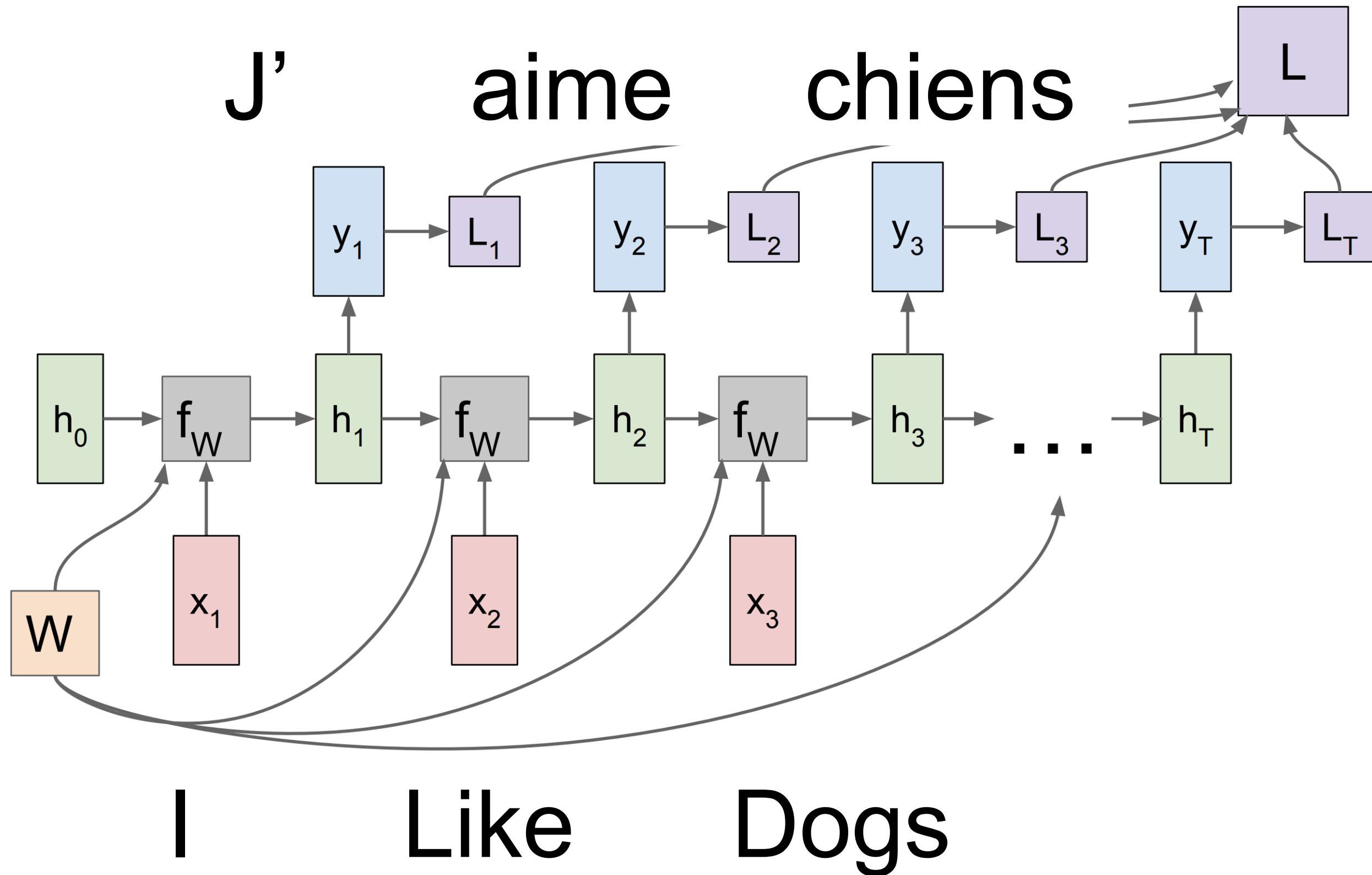


Using an RNN

RNN: Computational Graph: Many to Many

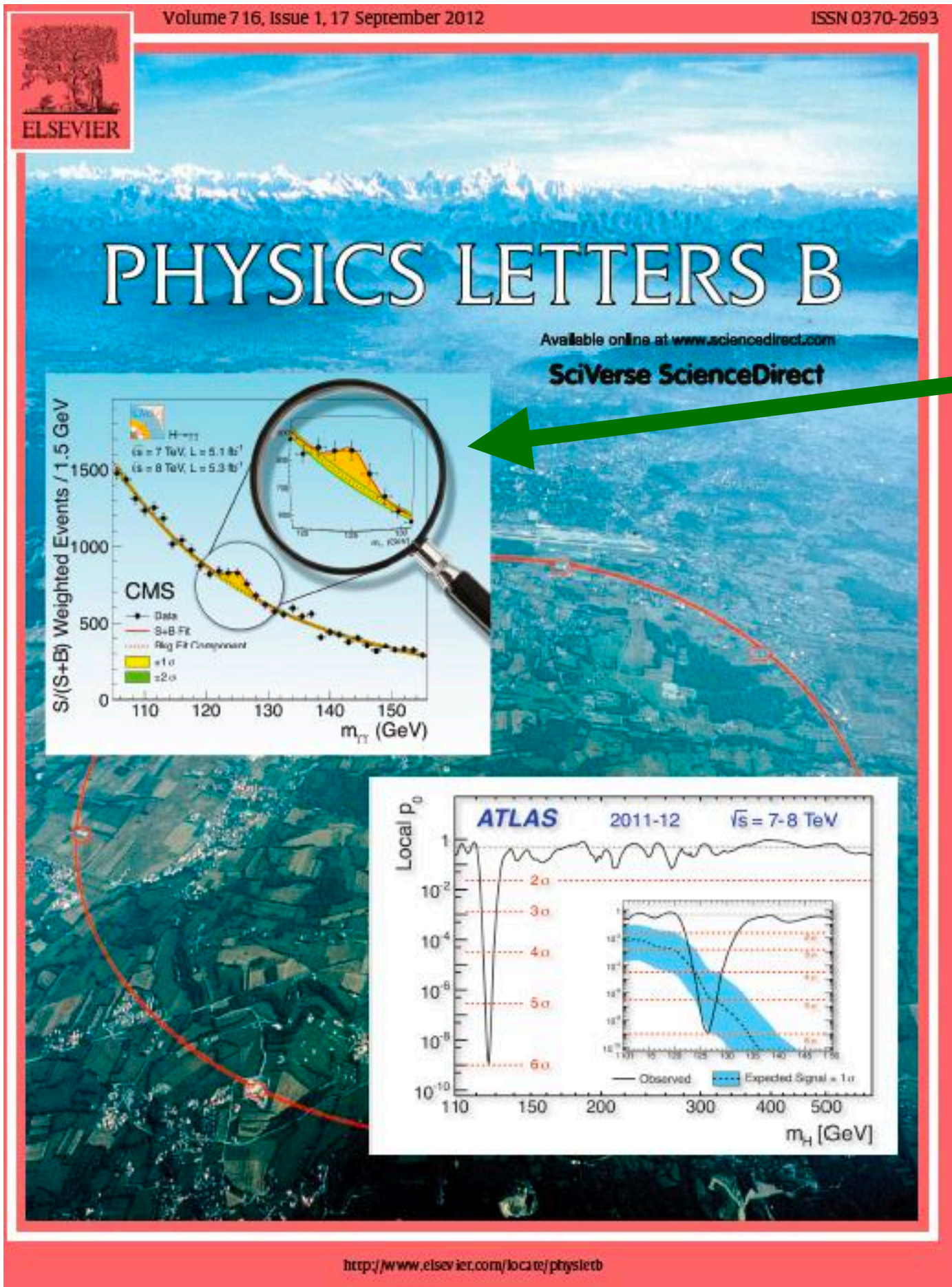


Using an RNN



A Point

That Plot has a photon energy NN regression



Summary

- This class we showed the flexibility of the NN
- The real insight here is that we modified the loss
- We tried to solve a problem different than classification
- **You can solve many more**

Bonus

Are you Hungry?

- Lets do something fun:
 - Online there is a recipe list of about 100k recipes
- Challenge:
 - Lets try to generate our own recipes
- Any ideas of how you can do this?